

Junior Cycle Maths

# Text & Tests

**First Year Maths**  
**Teacher's Manual**

**1**

**Paul Cooke • O.D. Morris • Deborah Crean**



The Celtic Press

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# Introduction

This manual is designed to help teachers to embrace the new Junior Certificate Maths Specification, and to aid in its correct implementation while using *Text and Tests 1* for First Year Maths students. It starts with a Sample Scheme of Work that can be used to plan the Units of Learning throughout the year.

Each chapter in the textbook has been designed to represent a Unit of Learning appropriate for a mixed level First Year class group. The specification has been analysed and the most appropriate Learning Outcomes have been selected to study. Each Learning Outcome has then been “Unpacked” into the four categories outlined by the Junior Cycle for Teachers. This shows a teacher what Knowledge, Skills, Understanding and Values a student should have, upon completion of the chapter.

These Learning Outcomes are then summarised in a mind map which has been created through the Lens of the Unifying Strand, with each Learning Outcome placed under one of the six Elements of the Unifying Strand. Following on from this, each chapter has a list of which of the 13 Learning Outcomes from the Unifying Strand are appropriate for that chapter or Unit of Learning.

Each chapter in the textbook has investigations spread throughout, along with assignments and problems at the end. This manual details possible answers and solutions, along with extra discussion points and extensions that teachers can use to help guide their students with these sections. These Learning Experiences are designed to be used throughout the chapter to enhance the learning of the students, rather than just at the end as a form of assessment.

We hope that teachers find this manual to be a helpful aid while teaching the new Junior Cycle Maths Specification.

Paul Cooke

O. D. Morris

Deborah Crean

## Scheme of Work

| Unit of learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Time allocation | Completed |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|
| 1. Natural numbers: <ul style="list-style-type: none"> <li>– Definition</li> <li>– Number Line</li> <li>– Place Value</li> <li>– Ascending/Descending</li> <li>– Prime and Composite Numbers</li> <li>– Product of Primes</li> <li>– Factor Trees</li> <li>– HCF/LCM</li> <li>– Squares/Square Roots</li> <li>– BOMDAS</li> <li>– Commutative/Associative/Distributive</li> <li>– Estimating</li> <li>– Significant Numbers</li> <li>– Word Problems</li> <li>– Assessment of Learning</li> </ul> |                 |           |
| TOTAL:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 |           |
| 2. Integers: <ul style="list-style-type: none"> <li>– Definition</li> <li>– Number Line</li> <li>– <math>&lt;</math> and <math>&gt;</math></li> <li>– Addition/Subtraction</li> <li>– Multiplication/Division</li> <li>– Brackets</li> <li>– Word Problems</li> <li>– Assessment of Learning</li> </ul>                                                                                                                                                                                           |                 |           |
| TOTAL:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 |           |



| Unit of learning                                                                                                                                                                                                                                                                                                                                                                 | Time allocation | Completed |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|
| <b>3. Fractions:</b> <ul style="list-style-type: none"> <li>– Equivalent Fractions</li> <li>– Fraction of a Given Number</li> <li>– Comparing Fractions</li> <li>– Addition/Subtraction</li> <li>– Multiplication/Division</li> <li>– Problems involving Fractions</li> <li>– Assessment of Learning</li> </ul>                                                                  |                 |           |
| <b>TOTAL:</b>                                                                                                                                                                                                                                                                                                                                                                    |                 |           |
| <b>4. Probability:</b> <ul style="list-style-type: none"> <li>– Listing Outcomes</li> <li>– Sample Spaces – Two-Way Tables</li> <li>– Fundamental Principle of Counting</li> <li>– Probability Scale</li> <li>– Words of Probability</li> <li>– Equally likely Outcomes</li> <li>– Calculating Probability</li> <li>– Word Problems</li> <li>– Assessment of Learning</li> </ul> |                 |           |
| <b>TOTAL:</b>                                                                                                                                                                                                                                                                                                                                                                    |                 |           |
| <b>5. Decimals:</b> <ul style="list-style-type: none"> <li>– Addition/Subtraction</li> <li>– Multiplication/Division</li> <li>– Ordering Decimals</li> <li>– Rounding and Estimates</li> <li>– Fractions to Decimals</li> <li>– Assessment of Learning</li> </ul>                                                                                                                |                 |           |
| <b>TOTAL:</b>                                                                                                                                                                                                                                                                                                                                                                    |                 |           |

| Unit of learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Time allocation | Completed |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|
| 6. Sets: <ul style="list-style-type: none"> <li>– List elements of a set</li> <li>– Clearly defined sets</li> <li>– Element / Not element</li> <li>– Sets defined by rules</li> <li>– Null Set</li> <li>– Equal Sets</li> <li>– Subsets (Proper/Improper)</li> <li>– Venn Diagrams</li> <li>– Union/Intersection</li> <li>– Cardinal Number</li> <li>– Universal Set</li> <li>– Complement of a Set</li> <li>– Sets and Probability</li> <li>– Word Problems</li> <li>– Assessment of Learning</li> </ul>                 |                 |           |
| TOTAL:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                 |           |
| 7. Percentages: <ul style="list-style-type: none"> <li>– Convert between Percentages, Fractions and Decimals</li> <li>– Find a percentage of a quantity</li> <li>– Express a number as a percentage of another number</li> <li>– Increase or Decrease a number by a given percentage</li> <li>– Find a number when given a percentage of it</li> <li>– Calculate VAT</li> <li>– Calculate discounts</li> <li>– Calculate Percentage Profit and Loss</li> <li>– Word Problems</li> <li>– Assessment of Learning</li> </ul> |                 |           |
| TOTAL:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                 |           |

| Unit of learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Time allocation | Completed |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|
| 8. Algebra: <ul style="list-style-type: none"> <li>– Use symbols to represent unknown numbers</li> <li>– Convert English words to Maths symbols</li> <li>– Understand like and unlike terms</li> <li>– Simplify Algebraic expressions</li> <li>– Substitute numbers for letters</li> <li>– Remove Brackets and Simplify</li> <li>– Multiply algebraic terms involving indices</li> <li>– Multiply brackets of the form <math>(x + a)(x + b)</math> where <math>a, b</math> are integers.</li> <li>– Use arrays to represent algebraic multiplication</li> <li>– Word Problems</li> <li>– Assessment of Learning</li> </ul> |                 |           |
| TOTAL:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 |           |
| 9. Area – Perimeter – Volume: <ul style="list-style-type: none"> <li>– Use the metric system and convert between different units of length</li> <li>– Calculate the perimeter of different 2D shapes made up of straight lines</li> <li>– Measure the area of rectangular and triangular 2D shapes</li> <li>– Find missing measurements when given the Perimeter or Area</li> <li>– Connect area of squares to square roots</li> <li>– Estimate area</li> </ul>                                                                                                                                                            |                 |           |

| Unit of learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Time allocation | Completed |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|
| <ul style="list-style-type: none"> <li>– Work with scaled drawings</li> <li>– Calculate the Volume and Surface Area of Cubes and Cuboids</li> <li>– Find missing measurements when given the Volume or Surface Area</li> <li>– Draw and visualize the Nets of Cubes and Cuboids</li> <li>– Word Problems</li> <li>– Assessment of Learning</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                 |           |
| TOTAL:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                 |           |
| <p>10. Geometry 1:</p> <ul style="list-style-type: none"> <li>– Recognise Points, Lines, Line Segments and Rays and understand the naming structure and symbols for each of these</li> <li>– Recognise different types of angles</li> <li>– Name angles using both a single letter and a 3-letter system</li> <li>– Understand the concept of a degree as a unit of measurement</li> <li>– Understand the terms axiom, proof and theorem</li> <li>– Measure and draw angles using a protractor</li> <li>– Recognise and calculate missing               <ul style="list-style-type: none"> <li>• Angles on a straight line</li> <li>• Angles at a point</li> <li>• Vertically opposite angles</li> <li>• Corresponding angles</li> <li>• Alternate angles</li> <li>• Interior angles</li> <li>• Supplementary angles</li> </ul> </li> </ul> |                 |           |

| Unit of learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Time allocation | Completed |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|
| <ul style="list-style-type: none"> <li>– Draw and explain parallel and perpendicular lines</li> <li>– Understand the meaning of collinear, transversal, revolution.</li> <li>– Word and visual problems</li> <li>– Assessment of Learning</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                |                 |           |
| TOTAL:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |           |
| <b>11. Ratio and Proportion:</b> <ul style="list-style-type: none"> <li>– Understand the notation of Ratio</li> <li>– Simplify ratios, including changing to same units</li> <li>– Divide a number in a given ratio</li> <li>– Understand the difference between ratio and proportion</li> <li>– Calculate the whole number when given a ratio or proportion of the number</li> <li>– Work with proportions in fractions, decimals and percentages</li> <li>– Direct Proportion</li> <li>– Unitary method</li> <li>– Use conversion graphs</li> <li>– Inverse Proportion</li> <li>– Best Buys</li> <li>– Word Problems</li> <li>– Assessment of Learning</li> </ul> |                 |           |
| TOTAL:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |           |
| <b>12. Statistics:</b> <ul style="list-style-type: none"> <li>– Collecting data</li> <li>– Surveys and Experiments</li> <li>– Types of data</li> <li>– Tally charts</li> <li>– Sampling and bias</li> <li>– Data Handling Cycle</li> <li>– Questionnaires</li> <li>– Assessment of Learning</li> </ul>                                                                                                                                                                                                                                                                                                                                                              |                 |           |
| TOTAL:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |           |

| Unit of learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Time allocation | Completed |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|
| 13. Coordinates: <ul style="list-style-type: none"> <li>– Fundamentals of coordinates (x,y)</li> <li>– Plotting and naming points</li> <li>– The four quadrants</li> <li>– Midpoint of a line segment</li> <li>– Slope of a Line               <ul style="list-style-type: none"> <li>i) Positive/negative</li> <li>ii) Rise/run</li> <li>iii) Formula</li> <li>iv) Word problems</li> </ul> </li> <li>– Assessment of Learning</li> </ul>                                                                                                                         |                 |           |
| TOTAL:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |           |
| 14. Solving equations: <ul style="list-style-type: none"> <li>– Understanding Equations</li> <li>– Forming Simple Equations</li> <li>– Solving Equations using the Balancing Method</li> <li>– Equations with letters on both sides</li> <li>– Equations with brackets</li> <li>– Word Problems</li> <li>– Assessment of Learning</li> </ul>                                                                                                                                                                                                                       |                 |           |
| TOTAL:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |           |
| 15. Geometry 2: <ul style="list-style-type: none"> <li>– Angles of a triangle</li> <li>– Exterior angle of a triangle</li> <li>– Special Triangles</li> <li>– Constructions               <ul style="list-style-type: none"> <li>• Bisector of an angle</li> <li>• Perpendicular bisector of a line segment</li> <li>• Line perpendicular to a given line</li> <li>• Line parallel to a given line</li> <li>• Division of a Line Segment into 3 sections</li> <li>• Draw a line segment of a given length</li> </ul> </li> <li>– Assessment of Learning</li> </ul> |                 |           |



| Unit of learning                                                                                                                                                                                                                                                                                                                                                 | Time allocation | Completed |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|
| TOTAL:                                                                                                                                                                                                                                                                                                                                                           |                 |           |
| 16. Presenting Data: <ul style="list-style-type: none"> <li>– Line Plots</li> <li>– Bar Charts</li> <li>– Frequency Tables</li> <li>– Stem and Leaf Plots</li> <li>– Assessment of Learning</li> </ul>                                                                                                                                                           |                 |           |
| TOTAL:                                                                                                                                                                                                                                                                                                                                                           |                 |           |
| 17. Transformation Geometry: <ul style="list-style-type: none"> <li>– Transformations               <ul style="list-style-type: none"> <li>• Translations</li> <li>• Axial Symmetry</li> <li>• Central Symmetry</li> </ul> </li> <li>– Symmetrical Shapes</li> <li>– Axis of Symmetry</li> <li>– Centre of Symmetry</li> <li>– Assessment of learning</li> </ul> |                 |           |
| TOTAL:                                                                                                                                                                                                                                                                                                                                                           |                 |           |



# Unit of Learning:

## 1. Natural Numbers

### Section A Unpacking the Learning Outcomes: \_\_\_\_\_

**N.1:** Students should be able to investigate the representation of numbers and arithmetic operations so that they can:

- represent the operations of addition, subtraction, multiplication, and division in  $\mathbb{N}$ , using models including the number line
- perform the operations of addition, subtraction, multiplication, and division and understand the relationship between these operations and the properties: commutative, associative and distributive in  $\mathbb{N}$

|                      |                                                                                                                                                                                                                                                                                                                           |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>that zero is not a natural number</li> <li>that addition and multiplication are both commutative and associative</li> <li>that subtraction and division are not commutative or associative</li> <li>the difference between an odd and even number</li> </ul> |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>add, subtract, multiply and divide natural numbers</li> <li>represent natural numbers on a number line</li> <li>list numbers in ascending and descending order</li> </ul>                                                                              |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>place value</li> <li>that certain operations should be done before others (BIRDMAS)</li> <li>that multiplication is distributive over addition</li> </ul>                                                                                              |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>the importance of natural numbers as the building blocks for all areas of mathematics</li> <li>the concept that natural numbers are infinite in a positive direction</li> </ul>                                                                        |

- explore numbers written as  $a^b$  (in index form) so that they can
  - flexibly translate between whole numbers and index representation of numbers
  - correctly use the order of arithmetic and index operations including the use of brackets

|                      |                                                                                                                                                                                                                                                                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>that <math>a^2</math> means <math>a</math> multiplied by <math>a</math></li> <li>the order in which index numbers come in the order of operations</li> </ul>                                        |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>convert between lists of repeated numbers and index notation<br/>e.g. <math>3 \times 3 \times 3 \times 3 \times 3 = 3^4</math></li> </ul>                                                     |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>that squaring a number and getting the square root of a number are inverse functions</li> <li>how to work with brackets and that, in expressions, they are always dealt with first</li> </ul> |

|              |                                                                                                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Value</b> | Students should appreciate: <ul style="list-style-type: none"> <li>the connection between the idea of a square number and a physical 2D square shape</li> </ul> |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|

- d) calculate and interpret factors (including the highest common factor), multiples (including the lowest common multiple), and prime numbers

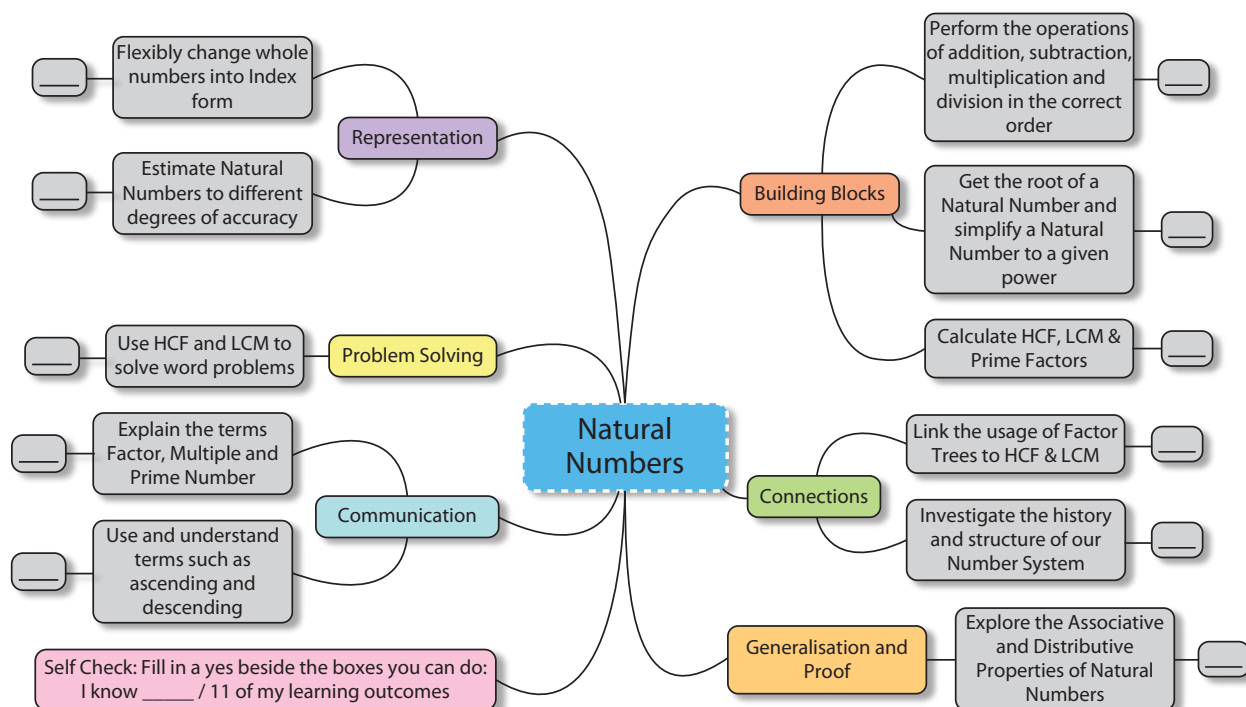
|                      |                                                                                                                                                                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>that a prime number only has two factors, itself and 1, and that if a number is not prime it must be composite</li> <li>how to write a number as a product of primes</li> </ul> |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>list the factors and multiples of any given number</li> <li>calculate the HCF or LCM of sets of numbers using lists or factor trees</li> </ul>                            |
| <b>Understanding</b> | Students should understand that: <ul style="list-style-type: none"> <li>2 is the only even prime factor</li> <li>HCF and LCM techniques can be applied to a wide range of word problems</li> </ul>                                           |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>the skills they develop as they become procedurally fluent, flexible, efficient and accurate</li> </ul>                                                                   |

- e) present numerical answers to the degree of accuracy specified, for example, correct to the nearest hundred or to three significant figures

|                      |                                                                                                                                                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know that: <ul style="list-style-type: none"> <li>if a number ends in a 0, 1, 2, 3, 4 the number rounds down</li> <li>if the number ends in a 5, 6, 7, 8, 9 the number rounds up</li> </ul>          |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>round natural numbers to the nearest 10, 100, 1000, etc.</li> <li>round a natural number to a given number of significant figures</li> </ul>      |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>that rounding to the nearest 10s/100s etc. is different to rounding to significant figures</li> <li>when it is appropriate to use each</li> </ul> |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>the usefulness of rounding numbers in real life, such as estimating bills etc.</li> </ul>                                                         |

## Section B Connecting the Unifying Strand:

All of these Learning Outcomes are summarised through the Lens of the Unifying Strand in the Mind Map, to help guide the Learning Experiences of the students:



## The Main Learning Outcomes from the Unifying Strand featured in this Unit of Learning are:

Students should be able to:

- U.1** recall and demonstrate understanding of the fundamental concepts and procedures that underpin each strand
- U.2** apply the procedures associated with each strand accurately, effectively, and appropriately
- U.4** represent a mathematical situation in a variety of different ways, including: numerically, algebraically, graphically, physically, in words; and to interpret, analyse, and compare such representations
- U.5** make connections within and between strands
- U.6** make connections between mathematics and the real world
- U.7** make sense of a given problem, and if necessary mathematise a situation
- U.8** apply their knowledge and skills to solve a problem, including decomposing it into manageable parts and/or simplifying it using appropriate assumptions
- U.9** interpret their solution to a problem in terms of the original question
- U.13** communicate mathematics effectively: justify their reasoning, interpret their results, explain their conclusions, and use the language and notation of mathematics to express mathematical ideas precisely

## Section C Investigations:

### Investigation 1:

- i) No, 3 does not divide into 20 equally
- ii) 1 person – 20 sweets  
2 people – 10 sweets  
4 people – 5 sweets  
5 people – 4 sweets  
10 people – 2 sweets  
20 people – 1 sweet

### Investigation 2:

#### Discussion Points:

- i) What is a prime number?
- ii) Why is 2 the only even prime number?
- iii) Why are other even numbers not prime?
- iv) Is there a finite number of prime numbers?
- v) What is the largest prime number you can find?

## Section D Learning Experiences:

### Assignment 1:

This is designed to get students thinking about our number system and how it works. It will help students to see a connection with History, and how numbers were developed over different centuries by different civilisations.

#### Discussion Points:

- Advantages and disadvantages of each number system
- Are there still times we use number systems other than decimal numbers (Time/Degrees)

### Assignment 2:

Students will need to think about the structure of natural numbers, and how the operation of subtraction works.

They will need to come up with the idea of splitting the numbers into a 2-digit group and a three-digit group and how best to do this to get the required end result.

Best Solution:  $234 - 65$

### Problem 1:

- Fill the 5 l bucket and empty it into the 3 l bucket – leaving 2 l in the 5 l bucket.
  - Empty out the 3 l bucket.
  - Fill the 3 l bucket with the 2 l from the 5 l bucket – leaving 2 l in the 3 l bucket.
  - Fill the 5 l bucket and fill the remaining 1 l space in the 3 l bucket from the 5 l bucket.  
Leaving 4 l in the 5 l bucket.
- Or

- Fill the 3 l bucket and empty into the 5 l bucket.
- Fill the 3 l bucket again and empty the contents of the 3 l bucket into the 5 l bucket. – Leaving the 5 l bucket full and 1 l in the 3 l bucket.
- Pour away the contents of the 5 l bucket
- Pour the 1 l from the 3 l bucket into the 5 l bucket.
- Fill the 3 l bucket
- Empty the contents of the 3 l bucket into the 5 l bucket. Leaving **4 l** in the 5 l bucket.

### *Problem 2:*

Best Order to Use clues: 3, 4, 2, 1

Answer: 1331

## Section E Chapter review:

1. Whole positive numbers are called **natural** numbers.
2. A number that divides evenly into a second number with no remainder is called a **factor** of the second number.
3. A **prime** number has 2 factors only.
4. LCM stands for **lowest common multiple**.
5. HCF stands for **highest common factor**.
6. A number multiplied by itself to give a second number is called the **square root** of the second number.
7. 1, 4, 9, 16, 25, ... are called **square** numbers.
8. BIRDMAS stands for **Brackets, Indices, Roots, Division, Multiplication, Addition, Subtraction**.
9. If  $(a * b) = (b * a)$  then the operation  $*$  is said to be **commutative**.
10. If  $a * (b * c) = (a * b) * c$  then the operation  $*$  is said to be **associative**.
11. Brackets are used to **prioritise** an operation.
12. When making a rough estimate of a calculation we generally **round** whole numbers off to the nearest 10 or 100.
13. If we estimate 32 657 as 33 000 we are using the first two **significant figures**.

# Unit of Learning:

## 2. Integers

### Section A Unpacking the Learning Outcomes:

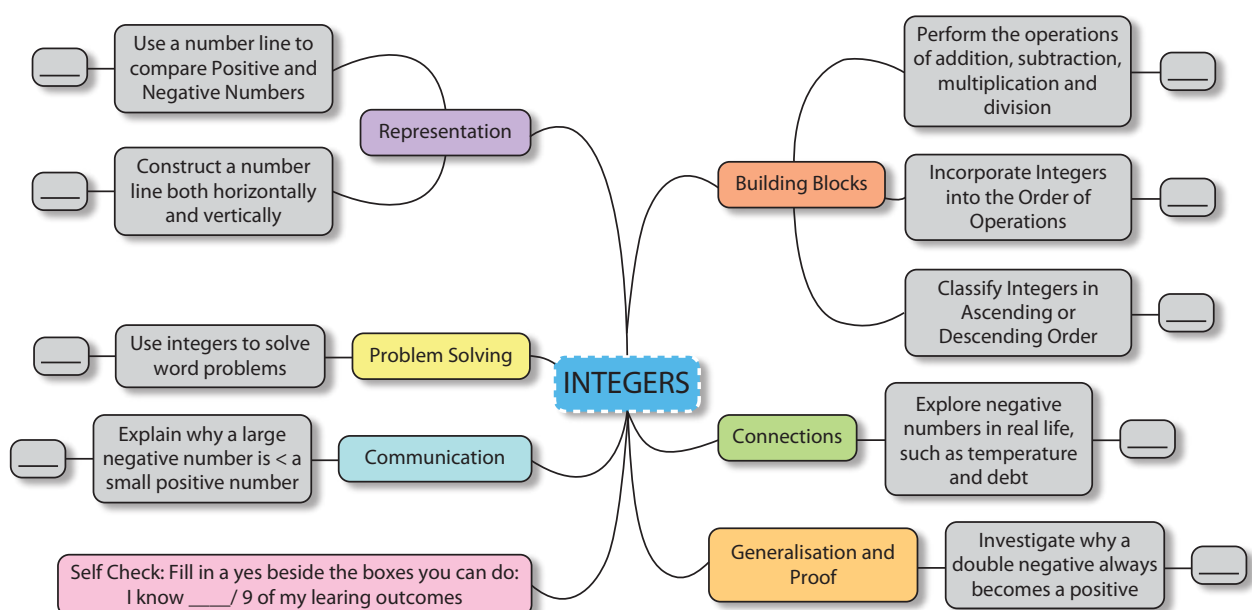
**N.1:** Students should be able to investigate the representation of numbers and arithmetic operations so that they can:

- represent the operations of addition, subtraction, multiplication, and division in  $\mathbb{Z}$ , using models including the number line
- perform the operations of addition, subtraction, multiplication, and division in  $\mathbb{Z}$

|                      |                                                                                                                                                                                                                                                                                             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>that zero is an integer</li> <li>the rules for signs when multiplying and dividing integers</li> <li>how to use <math>&lt;</math> and <math>&gt;</math> symbols in relation to integers e.g. <math>-4 &lt; 3</math></li> </ul> |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>add, subtract, multiply and divide integers</li> <li>represent integers on a number line</li> <li>list integers in ascending and descending order</li> </ul>                                                             |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>certain operations should be done before others (BIDMAS)</li> <li>the use of integers in problems involving temperature, money etc.</li> </ul>                                                                           |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>the importance of integers in the world around us</li> <li>the concept that integers are infinite in a positive and negative direction</li> </ul>                                                                        |

### Section B Connecting the Unifying Strand:

All of these Learning Outcomes are summarised through the Lens of the Unifying Strand in the Mind Map, to help guide the Learning Experiences of the students:





## The Main Learning Outcomes from the Unifying Strand featured in this Unit of Learning are:

Students should be able to:

- U.1** recall and demonstrate understanding of the fundamental concepts and procedures that underpin each strand
- U.2** apply the procedures associated with each strand accurately, effectively, and appropriately
- U.5** make connections within and between strands
- U.6** make connections between mathematics and the real world
- U.7** make sense of a given problem, and if necessary mathematise a situation
- U.8** apply their knowledge and skills to solve a problem, including decomposing it into manageable parts and/or simplifying it using appropriate assumptions
- U.9** interpret their solution to a problem in terms of the original question
- U.13** communicate mathematics effectively: justify their reasoning, interpret their results, explain their conclusions, and use the language and notation of mathematics to express mathematical ideas precisely

## Section C Investigations:

### *Investigation 1:*

#### Discussion Points:

- i) Applying integer number lines to a vertical problem
- ii) Working out a plan to tackle the problem as there are 2 components, where you end up and how many people are left in the lift at this stage
- iii) What other areas in real life might we need to use our addition and subtraction of integers skills?

Solutions: 1)  $2 - 3 + 1 - 2 = -2$

2) i)  $3 - 5 + 3 = 1$  (First floor)

ii)  $4 - 3 + 4 - 4 + 5 = 6$  (6 People still in lift)

### *Investigation 2:*

#### Discussion Points:

- i) Students should be able to discuss and explain the idea of integers as numbers with direction and how they move up and down a number line.
- ii) We are hoping to see students come to realise that a double negative will result in the number "moving" in the opposite direction i.e. becoming positive.
- iii) Students should be prompted to come up with ideas in real life where this occurs:
  - a. English – I'm not, not going to the party – I am going to the party
  - b. Finance – I take away some of the money you owe me – you now owe me less so the amount has become less negative

## Section D Learning Experiences:

### Problem 1:

- A. Students should notice that opposite numbers always add up to 7.
- B. i) Students should understand the concept of odd numbers and apply a negative sign to 1, 3 & 5.
- ii) Students should lay out different possibilities for how to add up all possible outcomes such as lists, grids, trees etc

### Discussion Points:

- Advantages and disadvantages of each way of laying out numbers.
- If a student has not already done so, you can display a sample space:

|    |    |    |    |    |     |    |
|----|----|----|----|----|-----|----|
| +  | −1 | 2  | −3 | 4  | −5  | 6  |
| −1 | −2 | 1  | −4 | 3  | −6  | 5  |
| 2  | 1  | 4  | −1 | 6  | −3  | 8  |
| −3 | −4 | −1 | −6 | 1  | −8  | 3  |
| 4  | 3  | 6  | 1  | 8  | −1  | 10 |
| −5 | −6 | −3 | −8 | −1 | −10 | 1  |
| 6  | 5  | 8  | 3  | 10 | 1   | 12 |

Solution: A 7 cannot be achieved.

- Discuss the possible combinations of numbers needed to make a 7 and how you need one odd number. These are all negative now, so it won't work etc.

### Assignment 2:

Students should recognise the use of mathematical skills in real world situations. This assignment will help to prepare them for doing mathematical research when needed, and to apply that research to a given question.

#### In the Black:

DEFINITION: Having positive net income; having greater income than expenses; making a profit.

Comes from the standard accounting practice of using black ink, as opposed to red ink, to denote positive values

#### In the Red:

DEFINITION: Having net losses; in debt.

Comes from the standard accounting practice of using red ink to denote negative values, especially a net loss.

#### Overdrawn:

DEFINITION: To draw money from (one's bank account) in excess of what the account holds.

A negative number is usually denoted with Dr, for Debit, and a positive number is usually denoted with Cr, for Credit.

*Problem 2:*

DEFINITION: Consecutive numbers are those that follow one directly after the other.

$$-4, -3, -2, -1, 0, 1, 2, 3, 4, 5$$

We need all but the number 5 to total to zero, so we need numbers to appear in positive/negative pairs. The only way this can happen is if we make 5 the largest number.

*Assignment 2:*

The students if not already aware, should discover the problem of dividing by 0, and the other properties of working with zero.

**Discussion Points:**

- If I have 5 sweets and decide to share them with nobody, but not keep them myself, where do the sweets go?  
(How do I Divide by zero)
- Why is it ok to have zero as the Numerator of a Fraction, but not the Denominator?
- Does zero perform the same what when dealing with both positive or negative numbers?
- Is zero prime/a natural number/an integer?

**Section E Chapter review:**

1. The set of positive and negative whole numbers including **zero** is called the set of integers.
2. When multiplying or dividing integers with like signs the answer is always **positive**.
3. When multiplying or dividing integers with different signs the answer is always **negative**.
4. Using an integer number line, numbers to the left are **less/smaller** than numbers on the right.
5. If the temperature is  $-5^{\circ}\text{C}$  it is often described as  $5^{\circ}\text{C}$  **below zero**.
6. If  $a$  and  $b$  are numbers and  $a > b$  then  $a$  is **greater than**  $b$ .
7. If  $c$  and  $d$  are numbers and  $c < d$  then  $c$  is **less than**  $d$ .
8. If  $e, f$ , and  $g$  are integer numbers and  $e > f$  and  $f > g$  then  $g$  is **less than**  $e$ .

# Unit of Learning:

## 3. Fractions

### Section A Unpacking the Learning Outcomes: \_\_\_\_\_

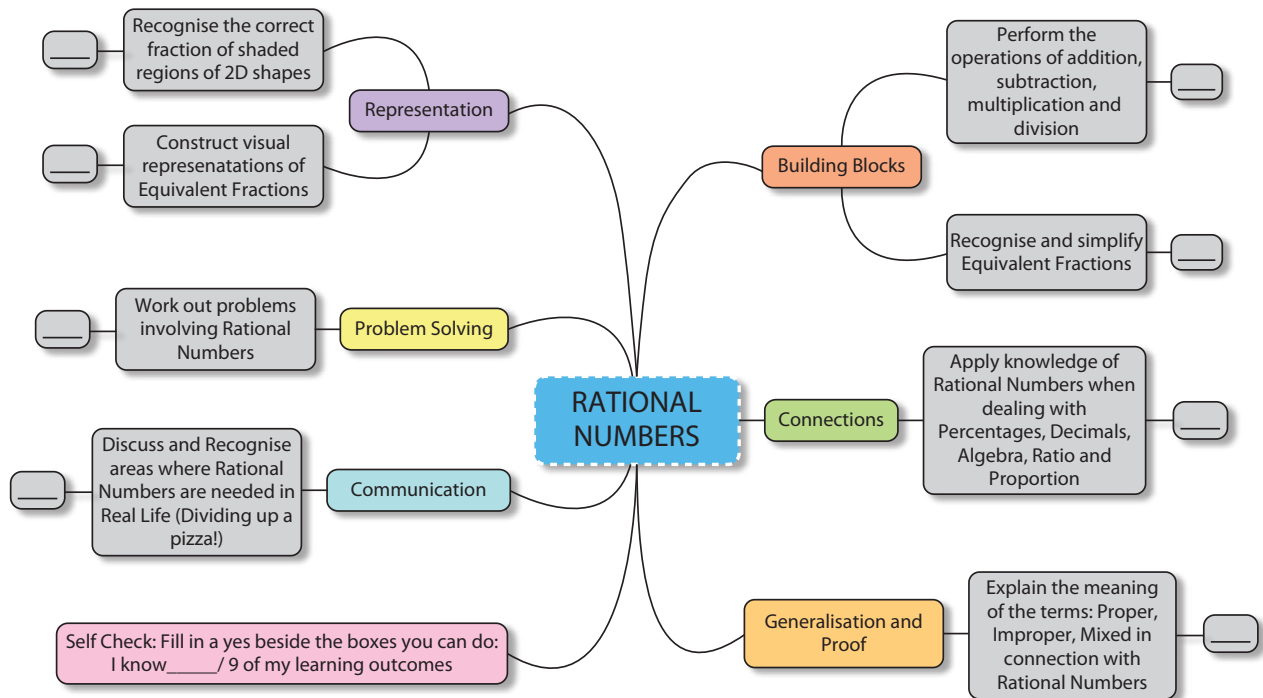
**N.1:** Students should be able to investigate the representation of numbers and arithmetic operations so that they can:

- represent the operations of addition, subtraction, multiplication and division in  $\mathbb{Q}$ , using models including the number line
- perform the operations of addition, subtraction, multiplication and division in  $\mathbb{Q}$

|                      |                                                                                                                                                                                                                                                                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | <p>Students should know:</p> <ul style="list-style-type: none"> <li>that the letter <math>\mathbb{Q}</math> stands for the set of rational numbers (fractions)</li> <li>the top of a fraction is the numerator and the bottom is the denominator</li> <li>how to find a fraction of a given number</li> <li>the meaning of LCM</li> </ul> |
| <b>Skills</b>        | <p>Students should be able to:</p> <ul style="list-style-type: none"> <li>use equivalent fractions to add and subtract</li> <li>multiply and divide fractions</li> <li>represent fractions on a number line</li> <li>list fractions in ascending and descending order</li> </ul>                                                          |
| <b>Understanding</b> | <p>Students should understand:</p> <ul style="list-style-type: none"> <li>equivalent fractions represent the same sized portion of a whole unit</li> <li>the words proper, improper and mixed in relation to fractions</li> <li>how to apply their fraction skills to word problems</li> </ul>                                            |
| <b>Value</b>         | <p>Students should appreciate:</p> <ul style="list-style-type: none"> <li>the concept of equivalent fractions and how we need them in real life e.g. Dividing into shares</li> <li>the logic of why we invert the second fraction and multiply, when we are dividing fractions</li> </ul>                                                 |

## Section B Connecting the Unifying Strand:

All of these Learning Outcomes are summarised through the Lens of the Unifying Strand in the Mind Map, to help guide the Learning Experiences of the students:



## The Main Learning Outcomes from the Unifying Strand featured in this Unit of Learning are:

Students should be able to:

- U.1** recall and demonstrate understanding of the fundamental concepts and procedures that underpin each strand
- U.2** apply the procedures associated with each strand accurately, effectively and appropriately
- U.3** recognise that equality is a relationship in which two mathematical expressions have the same value
- U.4** represent a mathematical situation in a variety of different ways, including: numerically, graphically, physically and in words
- U.5** make connections within and between strands
- U.6** make connections between mathematics and the real world
- U.7** make sense of a given problem, and if necessary mathematise a situation
- U.8** apply their knowledge and skills to solve a problem, including decomposing it into manageable parts and/or simplifying it using appropriate assumptions
- U.9** interpret their solution to a problem in terms of the original question
- U.13** communicate mathematics effectively: justify their reasoning, interpret their results, explain their conclusions, and use the language and notation of mathematics to express mathematical ideas precisely

## Section C Investigations:

### Investigation 1:

#### Discussion Points:

- i) Students should come up with the ideas that a and b should be members of the set of Integers ( $\mathbb{Z}$ ).
- ii) Discuss why a and b cannot themselves be fractions.
- iii) Following on from Assignment 2 in Chapter 2, students should be aware of the issues when we divide by zero. This should lead them to include the restriction that b cannot equal zero.

### Investigation 2:

The term fraction comes from the Latin term fractus, which means broken. We use the idea of fractions to represent portions of a whole, that are divided into equal sized pieces.

Fractures are the term given to pieces of material that are broken or cracked into pieces, so this in real life is a version of the word fraction.

## Section D Learning Experiences:

### Assignment 1:

- Students here will need to use lots of different research skills. They will need to measure heights accurately, and keep track of multiple drops happening, using tables/charts/diagrams etc
- They will need to compare two numbers and make one a fraction of the other.
- They may get different types of fractions and they will need to see if they are equivalent to each other.

#### Discussion Points:

- Connections with science:
  - a) Does the floor surface make a difference? Why?
  - b) Does the material/size of the ball make a difference? Why?
- Keeping track of results – what do students think is the best way?

### Problem 1:

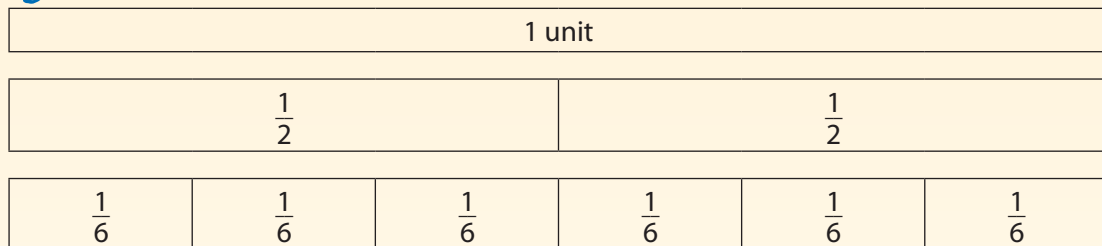
$$2^{\text{nd}} \text{ bounce: } \frac{1}{3} = 9 \text{ cm} \Rightarrow \frac{3}{3} = 27 \text{ cm}$$

$$1^{\text{st}} \text{ bounce: } \frac{1}{3} = 27 \text{ cm} \Rightarrow \frac{3}{3} = 81 \text{ cm}$$

#### Extension:

You could extend this problem for stronger students, asking them for the height of the ball after 3 or 4 bounces, and to draw a scaled diagram to represent the experiment.

### Assignment 2:



The students should notice that the problem of  $\frac{1}{2}$  divided by  $\frac{1}{6}$  is in fact asking them "How many  $\frac{1}{6}$  pieces are in one  $\frac{1}{2}$  piece?"

The above strips demonstrate the answer to this is 3.

### Problem 2:

Students will need to formulate a plan to figure out how much of the journey the passing of 5 minutes from the tower to the railway station represents.

They are looking for the difference between  $\frac{1}{4}$  and  $\frac{1}{3}$  of the journey. They could do this using fraction subtraction, or by using a fraction wall to see what size the piece is.

They should see that this 5-minute piece of the journey represents  $\frac{1}{12}$  of the journey.

Therefore, the full journey takes  $5 \times 12$  minutes = 60 minutes.

The question asked for the time he leaves and arrives. He has passed the clock tower after  $\frac{1}{4}$  of his journey. This is 15 mins, so we need to take 15 mins from 8:30; therefore he started his journey at 8:15 am.

He finishes his journey at 9:15 am, 60 minutes later.

## Section E Chapter review:

1. When any unit is broken into parts, **fractions** of the unit are created.
2. The bottom number of a fraction is called a **denominator**.
3.  $\frac{1}{2}, \frac{2}{4}, \frac{3}{6}$  are examples of **equivalent** fractions.
4. Fractions less than 1 are called **proper** fractions.
5. Improper fractions are fractions that are **greater** than 1.
6. To compare two fractions it is best to convert them to fractions with the same **denominator**.
7. The top number of a fraction is called a **numerator**.
8. If you are asked to find a fraction of a fraction you **multiply** the fractions.
9. Fractions are also called **rational numbers**.
10. To divide fractions, the second fraction must be **turned upside down** and multiplied by the first fraction.

# Unit of Learning:

## 4. Probability

### Section A Unpacking the Learning Outcomes: \_\_\_\_\_

**SP. 1:** Students should be able to investigate the outcomes of experiments so that they can:

- generate a sample space for an experiment in a systematic way, including tree diagrams for successive events and two-way tables for independent events
- use the fundamental principle of counting to solve authentic problems

|                      |                                                                                                                                                                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>the concept of a sample space</li> <li>if event 1 has <math>n</math> possible outcomes and event 2 has <math>m</math> possible outcomes, then the 2 events combined have <math>n \times m</math> possible outcomes</li> </ul> |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>list all possible outcomes of an event</li> <li>draw a two-way table to show all outcomes from 2 events</li> <li>apply the Fundamental Principle of Counting to word problems</li> </ul>                                |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>that the Fundamental Principle of Counting is a way of calculating the total number of outcomes from multiple events</li> <li>the terms experiment, event and outcome as used in probability</li> </ul>                 |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>that in order to work out the chance of a particular event happening, we must know all of the possible outcomes</li> <li>the usefulness of using tables instead of lists for multiple events</li> </ul>                 |

**SP. 2:** Students should be able to investigate random events so that they can:

- demonstrate understanding that probability is a measure on a scale of 0–1 of how likely an event (including an everyday event) is to occur
- use the principle that, in the case of equally likely outcomes, the probability of an event is given by the number of outcomes of interest divided by the total number of outcomes

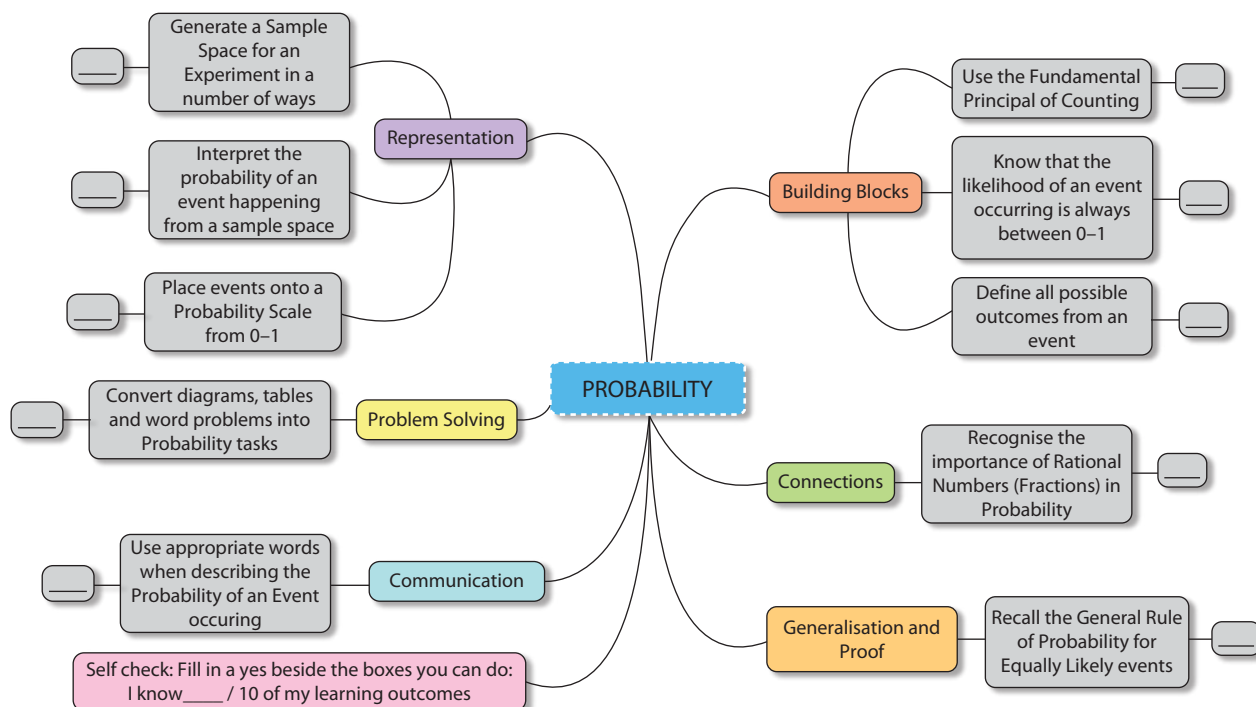
|                  |                                                                                                                                                                                                                                                                                                                           |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b> | Students should know: <ul style="list-style-type: none"> <li>how to place probabilities in the correct position on a scale</li> <li>that probabilities can be represented as fractions, decimals or percentages</li> <li>the structure of a standard deck of cards</li> </ul>                                             |
| <b>Skills</b>    | Students should be able to: <ul style="list-style-type: none"> <li>use appropriate words to describe the probability of an event, ranging from certain to impossible</li> <li>calculate the probability of an event using the rule: the number of outcomes of interest divided by the total number of outcomes</li> </ul> |



|                      |                                                                                                                                                                                                                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>• why the probability of an event occurring can only ever be between 0–1</li> <li>• the concept of equally likely outcomes</li> <li>• the notation of <math>P(\dots)</math></li> <li>• the terms Fair and Biased</li> </ul>         |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>• the study of Probability in terms of its connections to other areas of mathematics, such as fractions and sets</li> <li>• that the study of Probability can be applied to real life situations such as games of chance</li> </ul> |

## Section B Connecting the Unifying Strand:

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- U.9** interpret their solution to a problem in terms of the original question
- U.13** communicate mathematics effectively: justify their reasoning, interpret their results, explain their conclusions, and use the language and notation of mathematics to express mathematical ideas precisely

## Section C Investigations:

### Investigation 1:

- Each of these keys has 3 letters, so there are 3 outcomes each time. By the Fundamental Principal of Counting, there are  $3 \times 3 \times 3 \times 3 = 81$  possible outcomes from pressing these keys.
- Chart:

|                   | 4     | 6     | 6     | 3     | P                    |
|-------------------|-------|-------|-------|-------|----------------------|
| letters           | G H I | M N O | M N O | D E F |                      |
| Number of letters | 3     | 3     | 3     | 3     |                      |
| Word 1            | H     | O     | O     | D     | $2 + 3 + 3 + 1 = 9$  |
| Word 2            | G     | O     | O     | D     | $1 + 3 + 3 + 1 = 8$  |
| Word 3            | H     | O     | O     | F     | $2 + 3 + 3 + 3 = 11$ |

- By Pressing the No. 9 key repeatedly, you would get 48 three-letter variations:  
 $4 \times 4 \times 4 = 48$

### Investigation 2:

- Allow students time to come up with lots of different events and outcomes. They could do "Think – Pair – Share" to come up with ideas.
- Try to get students to firstly describe in words the probability of each event and then put some possible numbers on the outcomes.
- This could lead to a discussion about the fact that all possible outcomes for a given event must total to 1.

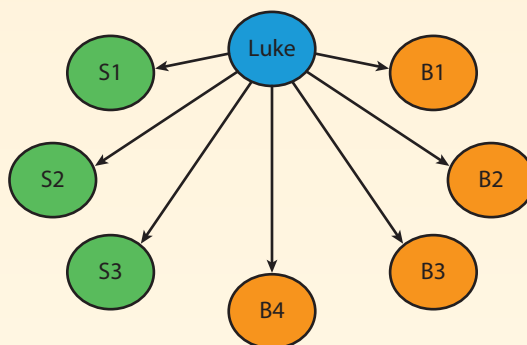
### Investigation 3:

EXAMPLE:

- |                    |                                                     |
|--------------------|-----------------------------------------------------|
| a. Impossible      | My teacher will be younger than me                  |
| b. Highly Unlikely | I will get to watch TV in all of my classes         |
| c. Unlikely        | We will be given extra time for lunch               |
| d. Evens           | The next student to walk in the door will be female |
| e. Likely          | I will get homework in some of my classes           |
| f. Highly Likely   | I will eat some food at lunch time                  |
| g. Certain         | The school bell will ring at the end of the day     |
- These types of exercises will often lead to discussions regarding the idea of things being impossible/certain. The students can have a discussion about these words and how we use them in everyday life.

## Section D Learning Experiences:

### Assignment 1:



Example above shows Luke buying 7 eggs. This is repeated by the 7 others. Resulting in  $8 \times 7 = 56$

### Assignment 2:

- Discuss the terms Expected vs experimental probability.
- Discuss the idea of 100 rolls, and how they think they could get the experimental closer to the expected probability.
- Discuss the idea of biased vs fair.

### Assignment 3:

The dictionary definition of "the Odds":

- the ratio between the amounts staked by the parties to a bet, based on the expected probability either way.
- the chances or likelihood of something happening or being the case.

When *gamblers* speak of odds they have a slightly different take on it with regards to probability.

The odds of 2 to 1 means: There are 2 favourable outcomes to 1 unfavourable outcome. A total of 3 outcomes.

In probability terms "2 to 1" converts to a probability of  $\frac{2}{3}$  or 66.6%

The "odds" of rolling a 6 are "1 to 5" (1 : 5) as opposed to the probability of rolling a 6 which is  $\frac{1}{6}$  (converting ratios, 1 : 5 to proportions,  $\frac{1}{6}, \frac{5}{6}$ )

### Problem 1:

3, 4, 0, 0, 3, 2, 4, 1, 2, 1

- The other 3 & 4 are the first to be placed.
- The 2 zeros are the only numbers that can go between the 4 and three as there are 2 empty spaces and the 1s or 2s cannot go here.
- Once this is done, the 1s and 2s are placed by process of elimination.

### Problem 2:

If there are 25 different birthdays, that leaves 5 other pupils. If we assume that all 5 of these share a birthday with one of the other 25 students, that makes 6 students with the same birthday.

## Section E Chapter review:

1. The results of a trial or experiment are called **outcomes**.
2. If one event has  $m$  possible outcomes and a second event has  $n$  possible outcomes, the two events have  $m \times n$  possible outcomes.
3. The result in the previous question is referred to as "The Fundamental Principle of **Counting**."
4. To show how likely/unlikely any event is to happen we use a **probability** scale.
5. The probability of an event that is certain is **1**.
6. If an event has a probability of 0 then the event is **impossible**.
7. If outcomes have the same chance (probability) of occurring, the outcomes are called **50–50** outcomes.
8. For equally likely outcomes, the probability of an event is equal to the number of **favourable** outcomes divided by the number of **possible** outcomes.
9. A **two-way** table shows the possible outcomes from two independent trials.
10. If an event has a fifty-fifty chance of occurring then the chance of it happening equals the chance of it **not happening**.

# Unit of Learning:

## 5. Decimals

### Section A Unpacking the Learning Outcomes: \_\_\_\_\_

**N.1:** Students should be able to investigate the representation of numbers and arithmetic operations so that they can:

- represent the operations of addition, subtraction, multiplication, and division in  $\mathbb{Q}$  using models including the number line, decomposition, and accumulating groups of equal size
- perform the operations of addition, subtraction, multiplication, and division and understand the relationship between these operations and the properties: commutative, associative and distributive in  $\mathbb{Q}$

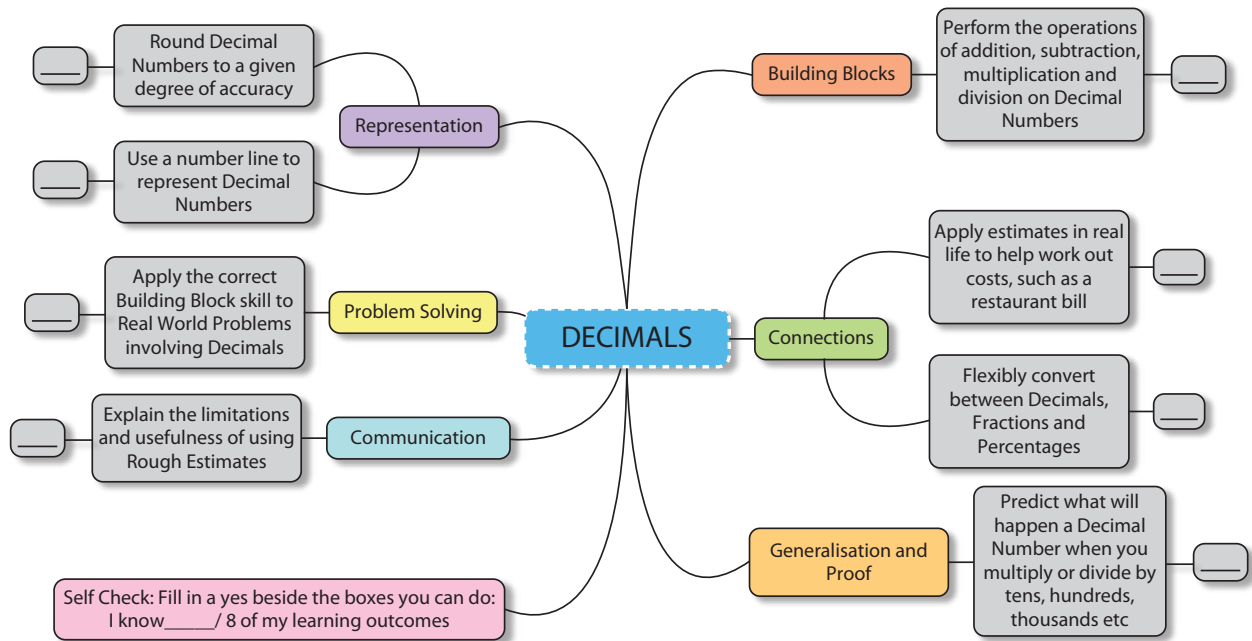
|                      |                                                                                                                                                                                                                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>how to convert between decimals and fractions, both by mathematical operations and by using a calculator</li> <li>how to use <math>&lt;</math> and <math>&gt;</math> symbols in relation to decimals eg <math>0.5 &lt; 0.6</math></li> </ul> |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>add, subtract, multiply and divide decimal numbers</li> <li>represent decimals on a number line</li> <li>list decimals in ascending and descending order</li> </ul>                                                                    |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>place value after the decimal point</li> <li>recurring decimals</li> <li>the relationship between decimals, fractions and percentages</li> </ul>                                                                                       |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>the importance of decimals in the world around us, especially in relation to money</li> </ul>                                                                                                                                          |

- present numerical answers to the degree of accuracy specified, for example, correct to two decimal places, or to three significant figures

|                      |                                                                                                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>that if a number ends in a 0, 1, 2, 3, 4 the number rounds down</li> <li>if the number ends in a 5, 6, 7, 8, 9 the number rounds up</li> </ul>                       |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>round decimals numbers correct to a given number of decimal places</li> <li>round a decimal number correct to a given number of significant figures</li> </ul> |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>that rounding to a given number of decimal places is different to rounding to significant figures</li> <li>when it is appropriate to use each</li> </ul>       |
| <b>Value</b>         | Students should: <ul style="list-style-type: none"> <li>value and appreciate the usefulness of rounding numbers in real life, such as estimating bills quickly etc.</li> </ul>                                                    |

## Section B Connecting the Unifying Strand:

All of these Learning Outcomes are summarised through the Lens of the Unifying Strand in the Mind Map, to help guide the Learning Experiences of the students:



## The Main Learning Outcomes from the Unifying Strand featured in this Unit of Learning are:

Students should be able to:

- U.1** recall and demonstrate understanding of the fundamental concepts and procedures that underpin each strand
- U.2** apply the procedures associated with each strand accurately, effectively, and appropriately
- U.3** recognise that equality is a relationship in which two mathematical expressions have the same value
- U.4** represent a mathematical situation in a variety of different ways, including: numerically, graphically, physically and in words
- U.5** make connections within and between strands
- U.6** make connections between mathematics and the real world
- U.7** make sense of a given problem, and if necessary mathematise a situation
- U.8** apply their knowledge and skills to solve a problem, including decomposing it into manageable parts and/or simplifying it using appropriate assumptions
- U.9** interpret their solution to a problem in terms of the original question
- U.13** communicate mathematics effectively: justify their reasoning, interpret their results, explain their conclusions, and use the language and notation of mathematics to express mathematical ideas precisely

## Section C Learning Experiences:

### Assignment 1:

- Here are lots of extra examples in English:  
deified, noon, civic, radar, level, rotor, kayak, reviver, racecar, redder, madam,  
Students could be asked to think of examples in English as a starter.
- $169.21 + 12.961 = 182.171$
- $182.171 + 171.281 = 353.452$
- $353.452 + 254.353 = 607.805$
- $607.805 + 508.706 = 1116.511$
- $1156.511 + 115.6111 = 1232.1221$
- $1232.1221 + 1221.2321 = \underline{\underline{2453.3542}}$
- It takes 6 reversals.

### Extension:

Question: How many numbers between 1 and 1000 are palindromes?

There is a pattern. There are exactly **10 palindromes** in each group of 100 numbers (after 99). Thus, there will be 9 sets of 10, or 90, plus the 18 from numbers 1 to 99, for a total of **108 palindromes** between 1 and 1,000.

### Problem 1:

Emma needs to have €2.50 in change.

- Emma's Change: 2 €1 coins, 2 20c coins, 2 5c coins.

If students quickly come up with the above combination, challenge them to explain why the 5 cent is the smallest coin she can have.

- Emma cannot have anything less than a 5c coin.  
If she had 2 2c the total would end in a 4  
If she had 2 1c the total would end in a 2  
If she had both the total would end in a 6  
So, none of these combinations would add up to a number that is a multiple of 10.

### Problem 2:

The number is 14.42

**Section D Chapter review:**

1. The first digit after the decimal point has **ten** as a denominator.
2. A quarter written as a decimal is **0.25**.
3. When adding or subtracting decimals (without using a calculator) always keep the decimal points **under** each other.
4. If a decimal number is multiplied by 100 the decimal point moves **two** places to the **right**.
5. If a decimal number is divided by 10 the decimal point moves **one** place to the **left**.
6. When rounding a decimal number correct to three places of decimals, if the fourth digit after the decimal point is 4 or less you should **leave** the third digit as it is.
7. If an event has a 50–50 chance of occurring then its probability as a decimal number is **0.5**.
8. In the decimal number 4.162 the 2 represents **two thousandths**.
9. When rounding a decimal number correct to two places of decimals, if the third digit after the decimal point is 5 or more you should increase the **second** digit by 1.
10. Using a calculator the **S<->D** key changes fractions to decimals and decimals to fractions.
11. Decimal numbers such as 42.3333... are called **recurring** decimals.
12. Decimal numbers such as 42.3333... can be written as **42. $\dot{3}$** .



# Unit of Learning:

## 6. Sets

### Section A Unpacking the Learning Outcomes: \_\_\_\_\_

**N.5:** Students should be able to explore the concept of a set so that they can:

- understand the concept of a set as a well-defined collection of elements, and that set equality is a relationship where two sets have the same elements
- define sets by listing their elements, if finite (including in a 2-set Venn Diagram), or by generating rules that define them

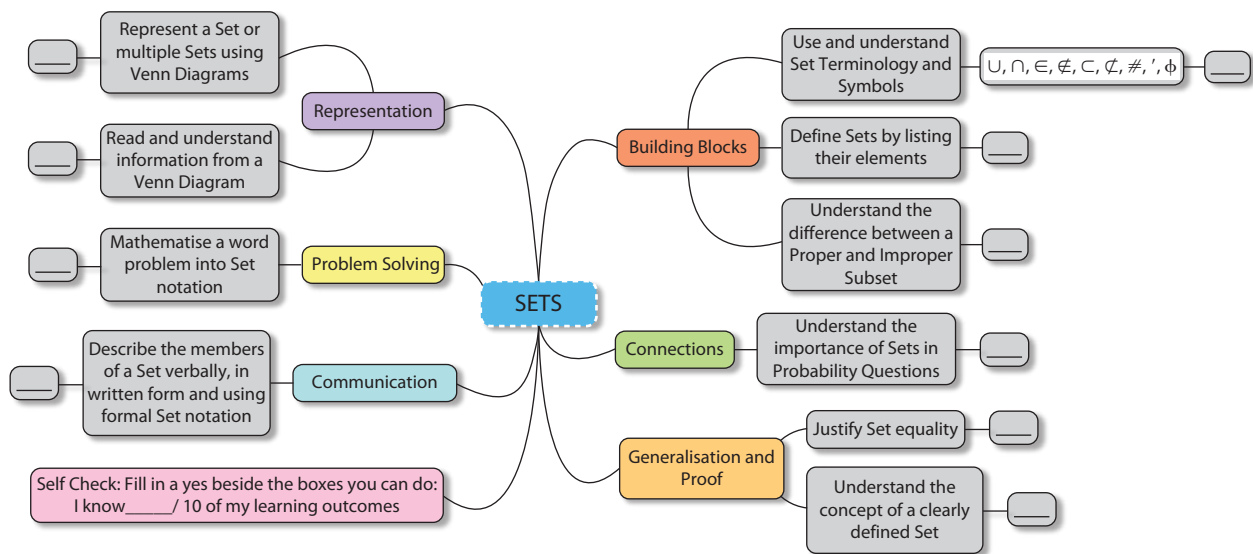
|                      |                                                                                                                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>how to create a rule to describe a set, and list elements when given a rule</li> <li>that the Null Set contains no elements</li> </ul> |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>list all of the elements of a set using Chain Brackets</li> <li>draw Venn Diagrams to define one or two Sets</li> </ul>          |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>that sets need to be <i>clearly defined</i></li> <li>the difference between a finite and infinite set</li> </ul>                 |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>that 2 sets are equal if they contain the same elements, even if they are listed in different orders</li> </ul>                  |

- use and understand suitable set notation and terminology, including null set,  $\emptyset$ , subset,  $\subset$ , complement, universal set, cardinal number,  $\#$ , intersection,  $\cap$ , union,  $\cup$
- perform the operations of intersection and union on two sets, including the use of brackets to define the order of operations

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>how to draw a Venn diagram to represent one or two sets</li> <li>the difference between a proper and improper subset</li> <li>what a universal set is</li> </ul>                                                                                                                                                                                    |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>identify the union and intersection of two sets from both lists of elements and Venn diagrams</li> <li>list a given number of subsets from a set</li> <li>shade in given regions on a Venn diagram or name regions using the correct notation</li> </ul>                                                                                      |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>the meaning of each of the symbols <math>\in</math>, <math>\notin</math>, <math>\subset</math>, <math>\not\subset</math>, <math>\cup</math>, <math>\cap</math>, <math>\emptyset</math>, <math>'</math>, <math>\#</math></li> <li>how brackets help to define the order of operations in sets</li> <li>the cardinal number of a set</li> </ul> |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>the connection between sets and probability and view sets as a useful technique to help solve probability problems</li> </ul>                                                                                                                                                                                                                 |

## Section B Connecting the Unifying Strand:

All of these Learning Outcomes are summarised through the Lens of the Unifying Strand in the Mind Map, to help guide the Learning Experiences of the students:



## The Main Learning Outcomes from the Unifying Strand featured in this Unit of Learning are:

Students should be able to:

- U.1** recall and demonstrate understanding of the fundamental concepts and procedures that underpin each strand
- U.2** apply the procedures associated with each strand accurately, effectively, and appropriately
- U.3** recognise that equality is a relationship in which two mathematical expressions have the same value
- U.4** represent a mathematical situation in a variety of different ways, including: numerically, graphically, physically and in words
- U.5** make connections within and between strands
- U.6** make connections between mathematics and the real world
- U.7** make sense of a given problem, and if necessary mathematise a situation
- U.8** apply their knowledge and skills to solve a problem, including decomposing it into manageable parts and/or simplifying it using appropriate assumptions
- U.9** interpret their solution to a problem in terms of the original question
- U.10** evaluate different possible solutions to a problem, including evaluating the reasonableness of the solutions, and exploring possible improvements and/or limitations of the solutions (if any)
- U.13** communicate mathematics effectively: justify their reasoning, interpret their results, explain their conclusions, and use the language and notation of mathematics to express mathematical ideas precisely

## Section C Investigations:

### Investigation 1:

- Students may start off the discussion believing that negative numbers cannot be prime using the argument that a Prime number can only have two factors, itself and 1. However, if we look at a prime number such as 3:
  - $1 \times 3 = 3$
  - $-1 \times -3 = 3$
- This shows us that in reality each prime number has in fact two pairs of factor, the positive and negative pairs.
- Following this logic, we must accept that any negative version of a Prime number is also prime, eg  $-3$ 
  - $-1 \times 3 = -3$
  - $3 \times -1 = -3$
- However, in the world of Mathematics there is an argument as to the validity of a negative number being prime. Some definitions of Prime numbers include the premise that they are only positive numbers, therefore ruling out negative prime numbers. Others argue, that negative primes are valid.
- If we only allow positive Prime numbers then they belong to the Natural Numbers set.
- If we allow negative Prime numbers, they belong to the Set of Integers.

### Investigation 2:

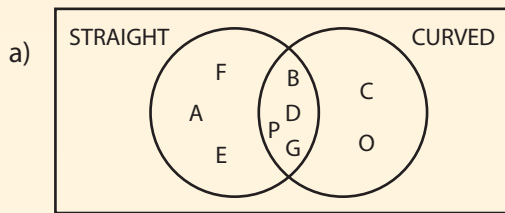
- i)  $\{\}, \{2, 3\}, \{2\}, \{3\}$  4 subsets
- ii)  $\{\}, \{2, 3, 4\}, \{2\}, \{3\}, \{4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}$  8 subsets
- iii)  $\{\}, \{2, 3, 4, 5\}, \{2\}, \{3\}, \{4\}, \{5\}, \{2, 3\}, \{2, 4\}, \{2, 5\}, \{3, 4\}, \{3, 5\}, \{4, 5\}, \{2, 3, 4\}, \{2, 3, 5\}, \{2, 4, 5\}, \{3, 4, 5\}$  16 subsets
- iv) Pattern: 4, 8, 16, ... numbers are doubling
- v)  $16 \times 2 = 32$  subsets in a 5-element set
- vi)  $32 \times 2 = 64$  subsets in a 6-element set
- vii) The number of subsets can be written as  $2^n$ , where  $n$  stands for the number of elements in the set.
  - E.g.  $2^6 = 64$ , for a set with 6 elements.

### Investigation 3:

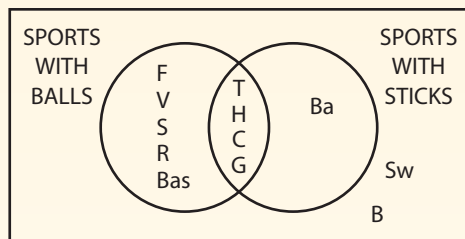
- a) Using her notation, the HCF can be found as the product of the elements in the INTERSECTION
- b) Using her notation, the LCM can be found as the product of all of the elements in the UNION
  - i) Using the idea of taking each prime factor as an individual element of a set, helps to calculate the HCF and LCM, ie the numbers work.
  - ii) However, in Sets we learn that we never repeat a term, so her method is flawed. The idea of using sets to represent prime factors breaks the rules we have for sets, and so is not a good concept.

## Section D Learning Experiences:

### Assignment 1:



- b) i) Sports that use a ball:  
 {Tennis, Football, Volleyball, Hurling, Soccer, Cricket, Rugby, Basketball, Golf}  
 ii) Sports that use a bat/club/racket etc:  
 {Tennis, Hurling, Cricket, Badminton, Golf}  
 iii) iv) v)



### Problem 1:

- We need the square root of  $n$  to be bigger than 9 but less than 11.
- 9 squared is 81 and 11 squared is 121.
- However, we cannot use these, as then the difference would be  $-1$  and  $+1$ .
- As the set must be whole positive numbers, we use the range of numbers from 82 – 120.

$$A = \{82, 83, 84, \dots, 118, 119, 120\}$$

All of these numbers have a square root that is no more than  $+/- 1$  away from 10.

$$\#A = 39 \text{ (you include both 82 and 120)}$$

### Problem 2:

#### Examples:

|                                                |                                          |
|------------------------------------------------|------------------------------------------|
| $A = \{x x \text{ is a red card}\}$            | $P(\text{element of } A) = \frac{1}{2}$  |
| $B = \{x x \text{ is a picture card}\}$        | $P(\text{element of } B) = \frac{3}{13}$ |
| $C = \{x x \text{ is a card with a 4 on it}\}$ | $P(\text{element of } C) = \frac{1}{13}$ |
| $D = \{x x \text{ is a Black 4}\}$             | $P(\text{element of } D) = \frac{1}{26}$ |
| $E = \{x x \text{ is a number card}\}$         | $P(\text{element of } E) = \frac{9}{13}$ |

### Assignment 2:

#### Examples:

$$A = \{1, 2, 3, 4, 5, 6\} \quad B = \{4, 5, 6, 7, 8\} \quad C = \{4, 5, 9, 10\}$$

- a) i)  $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$   
 $B \cup A = \{4, 5, 6, 7, 8, 1, 2, 3\}$

The elements are the same and order does not matter so:

$$A \cup B = B \cup A$$

$$\text{ii) } A \cap B = \{4, 5, 6\}$$

$$B \cap A = \{4, 5, 6\}$$

The elements are the same so:

$$A \cap B = B \cap A$$

Union and Intersection are Commutative.

$$\begin{aligned} \text{b) i) } A \cup (B \cup C) &= A \cup \{4, 5, 6, 7, 8, 9, 10\} \\ &= \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} \end{aligned}$$

$$\begin{aligned} (A \cup B) \cup C &= \{1, 2, 3, 4, 5, 6, 7, 8\} \cup C \\ &= \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} \end{aligned}$$

The elements are the same and order does not matter so:

$$A \cup (B \cup C) = (A \cup B) \cup C$$

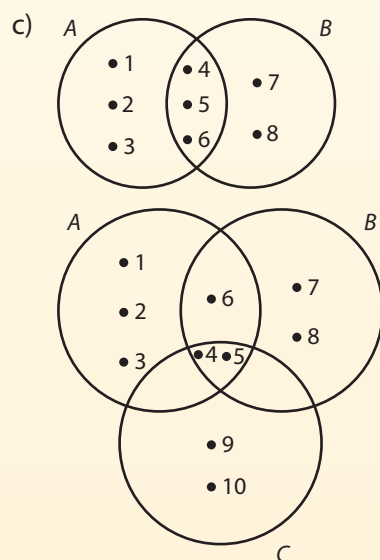
$$\begin{aligned} \text{ii) } A \cap (B \cap C) &= A \cap \{4, 5\} \\ &= \{4, 5\} \end{aligned}$$

$$\begin{aligned} (A \cap B) \cap C &= \{4, 5, 6\} \cap C \\ &= \{4, 5\} \end{aligned}$$

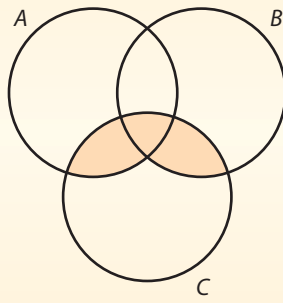
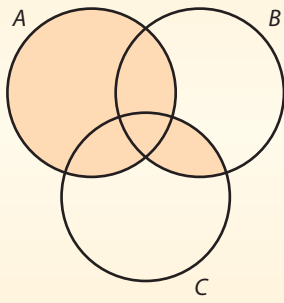
The elements are the same and order does not matter so:

$$A \cap (B \cap C) = (A \cap B) \cap C$$

Union and Intersection are Associative.



d) Is  $A \cup (B \cap C) = (A \cup B) \cap C$ ? **FALSE**



## Section E Chapter review:

1. The items in a set are called **elements**.
2. The elements of a set must be **clearly defined**.
3. Capital letters are used to **denote** a set.
4. No element in a set can be repeated more than **once**.
5. All the elements are enclosed in **chain brackets**.
6. Commas are used to **separate** elements.
7. Another name for an empty set is a **null** set.
8. The set  $B$  is a **subset** of set  $A$ , if every element of  $B$  is also an element of  $A$ .
9. A set itself and the null set are called **improper** subsets.
10. The **intersection** of two sets  $A$  and  $B$  is the set of elements that are in both  $A$  and  $B$ .
11. The union of two sets  $A$  and  $B$  is the set of all the elements in  $A$  and  $B$  without **repeating** any element.
12. The **cardinal** number of a set is the number of elements in the set.
13. The main set from which all other sets are taken is called the **universal** set and is denoted by  **$U$** .
14. Given a set  $A$ , the complement of  $A$  is the set of all the elements in  $U$  but **not in  $A$**  and is denoted by  **$A'$** .

# Unit of Learning:

## 7. Percentages

### Section A Unpacking the Learning Outcomes:

**N.2:** Students should be able to investigate equivalent representations of rational numbers so that they can:

a) flexibly convert between fractions, decimals, and percentages

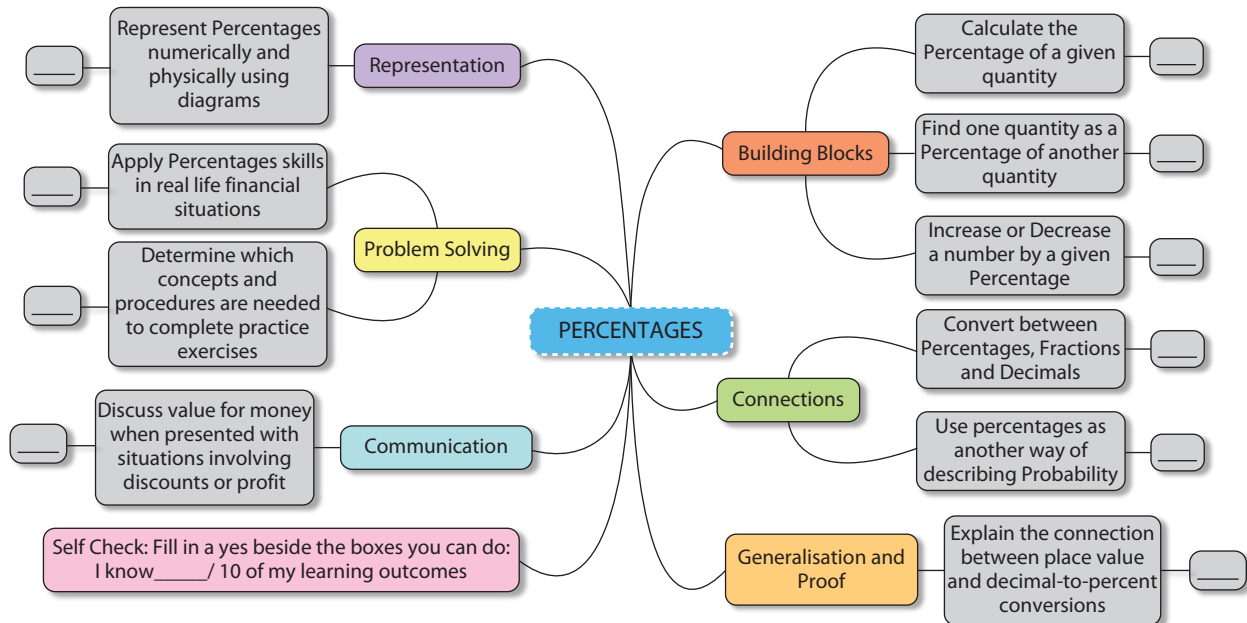
|                      |                                                                                                                                                                                                                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>to change a decimal to a percentage you multiply by 100</li> <li>to change a fraction to a percentage you multiply by <math>\frac{100}{1}</math></li> <li>to change a percentage to a decimal or fraction you divide by 100</li> </ul> |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>arrange a variety of number representations in ascending or descending order</li> <li>visually represent a percentage</li> <li>express one quantity as a percentage of another quantity</li> </ul>                               |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>that Percentages, Fractions and Decimals are all ways to represent a portion of a number</li> <li>how to calculate a percentage of a quantity</li> </ul>                                                                         |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>the importance and significance of percentages in the context of the real world and their lives generally</li> <li>the study of percentages in terms of their connections to other areas of mathematics</li> </ul>               |

c) solve money-related problems including those involving bills, VAT, profit or loss, % profit or loss (on the cost price), cost price, selling price

|                      |                                                                                                                                                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>how to find a number, given a percentage of it</li> <li>how to calculate the cost or selling price when presented with a profit as a percentage</li> </ul>            |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>increase or decrease a quantity by a given percentage</li> <li>present a number as a Percentage Profit or Loss</li> </ul>                                       |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>the meaning of the words VAT, Discount, Profit and Loss</li> <li>how to use their percentages skills to solve a wide range of money related problems</li> </ul> |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>the important role that percentages play in Financial Mathematics</li> <li>the use of percentages in retail businesses</li> </ul>                               |

## Section B Connecting the Unifying Strand:

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- U.9** interpret their solution to a problem in terms of the original question
- U.13** communicate mathematics effectively: justify their reasoning, interpret their results, explain their conclusions, and use the language and notation of mathematics to express mathematical ideas precisely



## Section C Investigations:

### *Investigation 1:*

- Make sure students are aware of all of the different names that sugar can be given in ingredients such as:

Here are a few examples:

1. Beet sugar
2. Blackstrap molasses
3. Brown sugar
4. Buttered syrup
5. Cane juice crystals
6. Cane sugar
7. Caramel
8. Carob syrup
9. Castor sugar

- Students could be asked to see who can find the most names for sugar.
- Students have been asked to compare the % sugar, so they need to make sure that they put the amount of sugar in the recipes over the total quantity of ingredients in the full recipe.
- This could lead to some rounding/estimating especially if you are dealing with solids/liquids etc.
- Discuss how it doesn't matter on the overall quantity of each recipe once you have calculated the % sugar.
- Discuss types of charts that could be used. This will indicate what charts they have studied at primary school and help guide the Presenting Data chapter later on.

### *Investigation 2:*

DEFINITION: A **percentile** (or a centile) is a measure used in statistics indicating the value below which a given percentage of observations in a group of observations fall. For example, the 20<sup>th</sup> **percentile** is the value (or score) below which 20% of the observations may be found.

- In sports or exams, percentiles can be used to compare a person's performance to the rest of the population or study group. Persons scoring above the 50<sup>th</sup> percentile would be classed as above average.
- In infant development, children's weight and height are measured on a regular basis. This allows medical professionals to monitor their growth compared to the rest of the population using percentiles.
- When comparing the class outcomes, students may discuss averages, guiding the learning for the statistics chapters later in the book. You can examine what measures of average/spread they are familiar with.

### Investigation 3:

#### Option 1: Take Discount First:

10% off means 90% left. 90% of €3750 = €3375

VAT @ 20% now needs to be added on, meaning you have 120% of €3375 = **€4050**

#### Option 2: Apply VAT First:

VAT @ 20% needs to be added on, meaning you have 120% of €3750 = €4500

We now take 10% off, meaning you have 90% of €4500 left. 90% of €4500 = **€4050**

**IT DOESN'T MATTER WHICH YOU DO FIRST.**

## Section D Learning Experiences:

### Assignment 1:

**There are currently four different rates of VAT in Ireland.**

- Standard rate of VAT – 23% – everything except those listed below
- Reduced rate of VAT – 13.5% – Goods and services including fuel, papers, etc
- Second reduced VAT rate of 9% – Mostly catering and entertainment
- Agricultural rate of VAT – 4.8% – farming etc
- Nil rate of VAT – 0% – exports etc

### Assignment 2:

i) A = 100%

B = 64% (since 20% (of 80%) was poured out we are left with 80% of 80%)

ii) 64% = 1152 ml

1% = 18 ml

C = 1800 ml (100%)

D = 1440 ml (80%)

### Problem 1:

25% of 60 = 15

S = 15

80% = 60

1% = 0.75

100% = 75

U = 75

80 as a percentage of 25 = 320%

M = 320

S + U + M = 15 + 75 + 320 = 410

*Problem 2:*

Let each bag = 100 g

Total mixture = 400 g

Wheat bran = 20% of 100 g = 20 g

Percentage volume of Wheat Bran =  $\frac{20}{400} = 5\%$

**Section E Chapter review:**

1. A percentage is a fraction with **100** as denominator.
2. Percentages can be changed into **fractions** and **decimals**.
3. **50** % is the same as a half.
4. When comparing quantities make sure that both quantities are in the same **units**.
5. If a quantity is decreased by 35% you now have **65** % of the original.
6. If a quantity is increased by 45% you now have **145** % of the original.
7. If goods are sold at less than cost price a **loss** is made.
8. VAT stands for **Value Added Tax**.
9. A **discount** is a percentage reduction in the selling price.
10. A percentage profit is calculated by dividing the profit by the **cost price** and then multiplying by 100%.

# Unit of Learning:

## 8. Algebra

### Section A Unpacking the Learning Outcomes:

**AF.2:** Students should be able to investigate situations in which letters stand for quantities that are variable so that they can:

- generate and interpret expressions in which letters stand for numbers
- find the value of expressions given the value of the variables

|                      |                                                                                                                                                                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>how to represent unknown numbers using symbols or letters</li> <li>that when an expression contains more than one operation, we follow BIRDMAS</li> </ul>                        |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>find the value of an expression by substituting in given numbers for letters</li> <li>translate an expression into an English statement and vice versa</li> </ul>          |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>the meaning of the words: variable, term, expression, like/unlike terms, constant and coefficient</li> </ul>                                                               |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>the use of letters in formulae they would have already come across in primary school such as area, volume etc</li> <li>the connection between the formulae used</li> </ul> |

**AF.3:** Students should be able to apply the properties of arithmetic operations to generate equivalent expressions so that they can develop and use appropriate strategies to:

- simplify
  - linear expressions in one or more variables with coefficients in  $\mathbb{Q}$
  - quadratic expressions in one variable with coefficients in  $\mathbb{Z}$
  - expressions of the form  $a / (bx + c)$ , where  $a, b, c \in \mathbb{Z}$
- multiply expressions of the form
  - $a(bx + cy + d)$ ;  $a(bx^2 + cx + d)$ ; and  $ax(bx^2 + cx + d)$ , where  $a, b, c, d \in \mathbb{Z}$
  - $(ax + b)(cx + d)$  where  $a, b, c, d \in \mathbb{Z}$

|                      |                                                                                                                                                                                                                                                                                                            |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>like and unlike terms, recognising that <math>x^2</math> and <math>x</math> are not like terms</li> <li>that when an expression contains more than one operation, we follow BIRDMAS</li> </ul>                                                |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>simplify an expression by grouping <i>like</i> terms together</li> <li>multiply single brackets by a constant or variable or both</li> <li>multiply 2 brackets, each with 2 terms, using an array model</li> </ul>                      |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>that the rules for multiplying or dividing integers apply to variables as well</li> <li>that they can apply their knowledge of indices when multiplying or dividing variables with indices. ie <math>(x^2)(x^3) = x^5</math></li> </ul> |

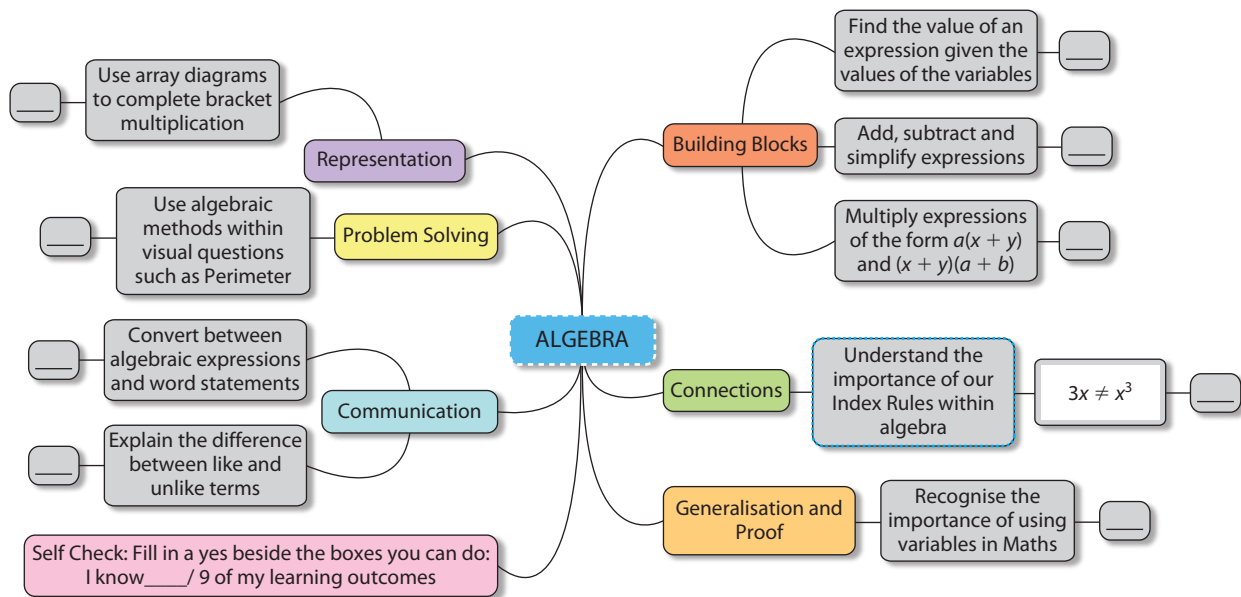
**Value**

Students should appreciate:

- that when we multiply or divide expressions a new version (usually simpler) of the original expressions is formed, and avoids repetition of like terms
- the connection between area and multiplication by examining array models

## Section B Connecting the Unifying Strand:

All of these Learning Outcomes are summarised through the Lens of the Unifying Strand in the Mind Map, to help guide the Learning Experiences of the students:



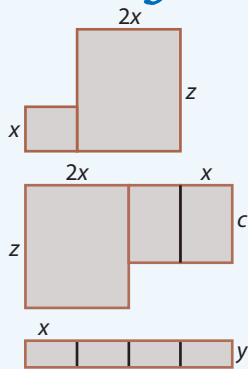
## The Main Learning Outcomes from the Unifying Strand featured in this Unit of Learning are:

Students should be able to:

- U.1** recall and demonstrate understanding of the fundamental concepts and procedures that underpin each strand
- U.2** apply the procedures associated with each strand accurately, effectively, and appropriately
- U.3** recognise that equality is a relationship in which two mathematical expressions have the same value
- U.4** represent a mathematical situation in a variety of different ways, including: numerically, graphically, physically and in words
- U.5** make connections within and between strands
- U.6** make connections between mathematics and the real world
- U.7** make sense of a given problem, and if necessary mathematise a situation
- U.8** apply their knowledge and skills to solve a problem, including decomposing it into manageable parts and/or simplifying it using appropriate assumptions
- U.9** interpret their solution to a problem in terms of the original question
- U.13** communicate mathematics effectively: justify their reasoning, interpret their results, explain their conclusions, and use the language and notation of mathematics to express mathematical ideas precisely

## Section C Investigations:

## Investigation 1:

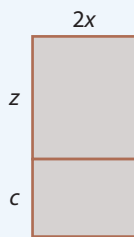
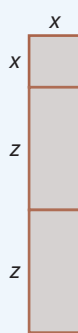


A

In order, top to bottom: a)  $x^2 + 2xz$       b)  $2xz + 2xc$       c)  $4xy$ 

B

Using approximately the same lengths for the sides we get:

d)  $2x(z + c)$ e)  $x(x + 2z)$ 

When you compare these to the given array models, students should notice that d) has the same area as b), and e) has the same area as c), they are just in a different arrangement.

This visually shows that:  $2x(z + c) = 2xz + 2xc$   
 $x(x + 2z) = x^2 + 2xz$

## Section D Learning Experiences:

## Assignment 1:

(i) to change  $4.2222 \dots$  to a fraction $4.2222 \dots = x$  (Let  $x$  equal the number you want to convert) $42.2222 \dots = 10x$  (Multiply both sides by 10) $4.2222 \dots = x$  (Subtract  $1x$  from  $10x$ ) $38.0 = 9x$  (The resulting number is worth  $9x$ ) $\therefore x = \frac{38}{9}$  (Divide both sides by 9) $\therefore \frac{38}{9} = 4.2222 \dots$  This is the value of  $4.2222 \dots$  as a fraction

(ii) to change 1.262626 ... to a fraction

$$1.2626 \dots = x \quad (\text{Let } x \text{ equal the number you want to convert})$$

$$126.2626 \dots = 100x \quad (\text{Multiply both sides by 100})$$

$$\begin{array}{r} 1.2626 \dots = x \\ 126.2626 \dots = 100x \\ \hline 125.0 \dots = 99x \end{array} \quad (\text{Subtract } 1x \text{ from } 100x)$$

$$125.0 \dots = 99x \quad (\text{The resulting number is worth } 99x)$$

$$\therefore x = \frac{125}{99} \quad (\text{Divide both sides by 99})$$

$$\therefore \frac{125}{99} = 1.262626 \dots \text{ This is the value of } 1.262626 \dots \text{ as a fraction}$$

(iii) to change 1.6666 ... to a fraction

$$1.6666 \dots = x \quad (\text{Let } x \text{ equal the number you want to convert})$$

$$16.6666 \dots = 10x \quad (\text{Multiply both sides by 10})$$

$$\begin{array}{r} 1.6666 \dots = x \\ 16.6666 \dots = 10x \\ \hline 15.0 \dots = 9x \end{array} \quad (\text{Subtract } 1x \text{ from } 10x)$$

$$15.0 \dots = 9x \quad (\text{The resulting number is worth } 9x)$$

$$\therefore x = \frac{15}{9} \quad (\text{Divide both sides by 9})$$

$$\therefore \frac{15}{9} = \frac{5}{3} = 1.6666 \dots \text{ This is the value of } 1.6666 \dots \text{ as a fraction}$$

## Assignment 2:

**Sir William Rowan Hamilton**, (4 August 1805 – 2 September 1865) is considered the greatest Irish Mathematician of all time and is best known for his discovery of Quaternions. For many years he had been trying to find a satisfactory way to multiply points in three dimensions, in such a way as to allow division. The idea of using four dimensions instead, and the way to do it, came to him in a flash, as he walked with his wife by the Royal Canal. Since it is one of the few major mathematical discoveries which is precisely located in time and circumstances, the event is very well-known in the international mathematical community, and people from all over the world know about "Hamilton's Bridge". When he made the discovery, Hamilton paused under Broom Bridge, took out his penknife, and scratched the fundamental formula:  $i^2 = j^2 = k^2 = ijk = -1$  into the stone. The day was the 16th of October, 1843, and there is now an annual walk from Dunsink Observatory, where Hamilton was working at the time, to Broom Bridge where there is a plaque commemorating the discovery. Quaternions are an essential mathematical tool for satellite navigation and space travel.

**Albert Einstein** (14 March 1879 – 18 April 1955) was a German-born theoretical physicist who developed the theory of relativity, one of the two pillars of modern physics (alongside quantum mechanics). His work is also known for its influence on the philosophy of science. He is best known by the general public for his mass–energy equivalence formula  $E = mc^2$  (which has been dubbed "the world's most famous equation"). He received the 1921 Nobel Prize in Physics "for his services to theoretical physics, and especially for his discovery of the law of the photoelectric effect", a pivotal step in the evolution of quantum theory.

**Al-Khwarizmi's** popularising treatise on algebra (*The Compendious Book on Calculation by Completion and Balancing*), ca. 813–833 CE presented the first systematic solution of linear and quadratic equations in Arabic. One of his principal achievements in algebra was his demonstration of how to solve quadratic equations by completing the square, for which he provided geometric justifications.

## Problem 1:

i)  $2^2 - 2 + 11 = 13$

ii)  $3^2 - 3 + 11 = 17$

iii)  $5^2 - 5 + 11 = 31$

iv)  $7^2 - 7 + 11 = 53$

v)  $9^2 - 9 + 11 = 83$

vi)  $11^2 - 11 + 11 = 121$  NOT PRIME

4 more:

- i)  $4^2 - 4 + 11 = 23$
- ii)  $6^2 - 6 + 11 = 41$
- iii)  $8^2 - 8 + 11 = 67$
- iv)  $10^2 - 10 + 11 = 101$

### Problem 2:

If we realise that each of the first two statements are giving us different ways of writing the letters  $c$  and  $a$ , we can use substitution for the final statement:

$$a + a + b = c + x + x$$

$$2a + b = c + 2x$$

(simplify)

$$2(b + x) + b = a + b + 2x$$

(substituting the first two statements)

$$2b + 2x + b = b + x + b + 2x$$

(simplify LHS and Substitute in RHS)

$$3b + 2x = 2b + 3x$$

$$3b + 2x - 2x = 2b + 3x - 2x$$

$$3b = 2b + x$$

$$3b - 2b = 2b - 2b + x$$

$$b = x$$

Or

$$a + a + b = c + x + x$$

$$(b + x) + (b + x) + b = (a + b) + x + x$$

$$(b + x) + (b + x) + b = ((b + x) + b) + x + x$$

$$b + x + b + x + b = b + x + b + x + x$$

$$3b + 2x = 2b + 3x$$

$$3b + 2x - 2x = 2b + 3x - 2x$$

$$3b = 2b + x$$

$$3b - 2b = 2b - 2b + x$$

$$b = x$$

## Section E Chapter review:

1. An **expression** is a collection of terms.
2. When simplifying expressions, we can add **like** terms.
3. Since the values of  $x$  and  $y$  are not known and since they can change value they are called **variables**.
4. The **coefficient** of  $x$  is the number of  $x$ s in an expression.
5. A term which has a fixed value is called a **constant** term.
6. To **evaluate** an expression, substitute numbers for variables.
7. To multiply brackets an **array** model can be used.



# Unit of Learning:

## 9. Area – Perimeter – Volume

### Section A Unpacking the Learning Outcomes:

**GT.1:** Students should be able to calculate, interpret, and apply units of measure

|                      |                                                                                                                                                                                                                               |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>• which unit of measurement mm/cm/m/km is the most appropriate to use on a given problem</li> </ul>                                                              |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>• convert between different units of measurement</li> <li>• use equipment such as a ruler, tape or trundle wheel to measure distances</li> </ul>           |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>• and be able to visualise each unit of measurement, and be able to make reasonable estimations of distances, such as the length of a room etc.</li> </ul> |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>• the importance of being able to measure lengths in real life and recognise this as an important life skill</li> </ul>                                    |

**GT.2:** Students should be able to investigate 2D shapes and 3D solids so that they can:

a) draw and interpret scaled diagrams

|                      |                                                                                                                                                                                                    |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>• how to pick a suitable scale when drawing a floor plan or other scaled diagram</li> </ul>                                           |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>• convert from a scaled diagram to get real life measurements</li> </ul>                                                        |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>• the method for converting real measurements into smaller units for a scaled diagram</li> </ul>                                |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>• how important it is to be able to use scaling in real life areas such as architecture, engineering, geography etc.</li> </ul> |

b) draw and interpret nets of rectangular solids

|                  |                                                                                                                                                                                                                |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b> | Students should know: <ul style="list-style-type: none"> <li>• the relationship between 2D shapes and 3D solids</li> <li>• how to draw a net from a 3D solid and recognise the set of nets of cubes</li> </ul> |
| <b>Skills</b>    | Students should be able to: <ul style="list-style-type: none"> <li>• physically manipulate 2D shapes (nets) to create 3D solids</li> <li>• visualise and cognitively manipulate 2D and 3D solids</li> </ul>    |

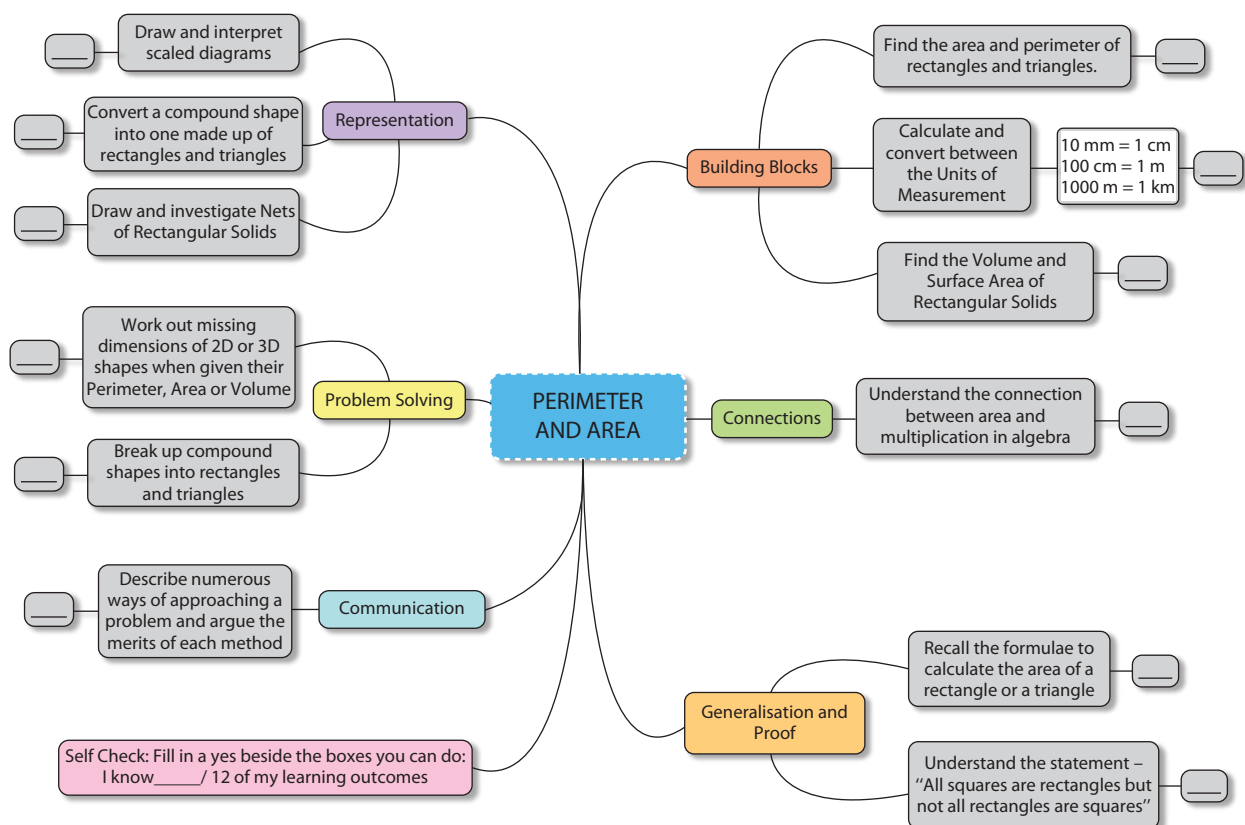
|                      |                                                                                                                                                                                                                                             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>that not all compound shapes form nets and make predictions as to whether they will or not</li> <li>the need to develop their ability to reason geometrically</li> </ul> |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>the importance and significance of 2D shapes and 3D solids in the context of the real world and their lives generally</li> </ul>                                         |

- c) find the perimeter and area of plane figures made from combinations of triangles and rectangles  
 d) & e) find the volume and surface area of rectangular solids

|                      |                                                                                                                                                                                                                    |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>the properties of specified 2D shapes and 3D solids</li> <li>how to calculate the volume of cuboids</li> </ul>                                        |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>calculate the perimeter and area of rectangular or triangular 2D shapes</li> <li>calculate the perimeter and area of compound shapes</li> </ul> |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>the connection between nets and surface area and be able to calculate surface area</li> </ul>                                                   |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>2D and 3D shapes in terms of their connections to other areas of mathematics</li> </ul>                                                         |

## Section B Connecting the Unifying Strand:

All of these Learning Outcomes are summarised through the Lens of the Unifying Strand in the Mind Map, to help guide the Learning Experiences of the students:



## The Main Learning Outcomes from the Unifying Strand featured in this Unit of Learning are:

Students should be able to:

- U.1** recall and demonstrate understanding of the fundamental concepts and procedures that underpin each strand
- U.2** apply the procedures associated with each strand accurately, effectively, and appropriately
- U.4** represent a mathematical situation in a variety of different ways, including: numerically, algebraically, graphically, physically, in words; and to interpret, analyse, and compare such representations
- U.5** make connections within and between strands
- U.6** make connections between mathematics and the real world
- U.7** make sense of a given problem, and if necessary mathematise a situation
- U.8** apply their knowledge and skills to solve a problem, including decomposing it into manageable parts and/or simplifying it using appropriate assumptions
- U.9** interpret their solution to a problem in terms of the original question
- U.11** generate general mathematical statements or conjectures based on specific instances
- U.13** communicate mathematics effectively: justify their reasoning, interpret their results explain their conclusions, and use the language and notation of mathematics to express mathematical ideas precisely

## Section C Investigations:

### *Investigation 1:*

- Students firstly need to understand that they have 10 m of fencing to make one length and one width, as they have 20 m in total.
- This investigation is meant to assess the level of knowledge of area before moving onto this topic in the next section.

Possible areas:

- $1 \times 9 = 9 \text{ m}^2$
- $2 \times 8 = 16 \text{ m}^2$
- $3 \times 7 = 21 \text{ m}^2$
- $4 \times 6 = 24 \text{ m}^2$
- $5 \times 5 = 25 \text{ m}^2$
- $6 \times 4 = 24 \text{ m}^2$
- $7 \times 3 = 21 \text{ m}^2$
- $8 \times 2 = 16 \text{ m}^2$
- $9 \times 1 = 9 \text{ m}^2$
- Largest area is  $25 \text{ m}^2$ , formed from a  $5 \times 5$  squares (4 sides of 5 m each total to 20 m of fencing)
- **Extension:**  
You could ask students to draw a graph of length vs area if students know about graphs from Primary School. This would introduce them to a Quadratic shaped curve.

### Investigation 2:

A: ABCD is a square of 8 cm so the area is:  $8 \times 8 = 64 \text{ cm}^2$

B:

#### Method 1:

- Each triangle has a base of 6 cm and a height of 2 cm so its area is:  $\frac{1}{2} (6) (2) = 6 \text{ cm}^2$
- There are 4 triangles so  $4 \times 6 = 24 \text{ cm}^2$
- Area of EFGH = Area of ABCD – 4 triangles  
 $= 64 - 24$   
 $= 40 \text{ cm}^2$

#### Method 2:

- You can count boxes and get an estimation, by combining partial boxes together. Each box represents  $1 \text{ cm}^2$

#### Method 3:

- You can divide up EFGH internally into a  $4 \times 4$  square and 4 triangles with base of 6 cm and a height of 2 cm  
 $\text{Area} = 4 \times 4 = 16 \text{ cm}^2$   
 $+ 4 (\frac{1}{2}(6)(2)) = 4(6) = 24 \text{ cm}^2$        $16 + 24 = 40 \text{ cm}^2$

C: % not covered =  $\frac{24}{64} = 0.375 = 37.5\%$

D: The internal square takes up 62.5% of the total area = 0.625 in decimal.

### Investigation 3:

#### Discussion points:

- Selecting a suitable scale
- Curved walls
- Non-regular shapes
- Drawing windows and doors on scaled diagrams
- Calculate area and perimeter of each room/total ground floor.

### Investigation 4:

A:

- 3 faces on each cube will be painted.
- $\frac{3}{6}$  of each cube's faces are painted, so 50% of each cube is painted.
- 50% of the surface area of all the blocks are painted.

B:

- $9 \times 3$  blocks are needed, 27 blocks in total.
- i) 3 faces painted: 8 blocks
- ii) 2 faces painted: 12 blocks
- iii) 1 face painted: 6 blocks
- (1 internal block not painted)

C:

|                 | 2 blocks along<br>each side (8 total) | 3 blocks along<br>each side (27 total) | 4 blocks along<br>each side (64 total) |
|-----------------|---------------------------------------|----------------------------------------|----------------------------------------|
| 1 face painted  | 0                                     | 6                                      | 24                                     |
| 2 faces painted | 0                                     | 12                                     | 24                                     |
| 3 faces painted | 8                                     | 8                                      | 8                                      |
| Internal        | 0                                     | 1                                      | 8                                      |

### Investigation 5:

i) Volume of box =  $1\text{ m} \times 1\text{ m} \times 1\text{ m} = 1\text{ m}^3$

OR

Volume of box =  $100\text{ cm} \times 100\text{ cm} \times 100\text{ cm} = 1,000,000\text{ cm}^3$

ii) Volume of sugar cube =  $0.01\text{ m} \times 0.01\text{ m} \times 0.01\text{ m} = 0.000001\text{ m}^3$

OR

Volume of sugar cube =  $1\text{ cm} \times 1\text{ cm} \times 1\text{ cm} = 1\text{ cm}^3$

a) 100 cubes in a row

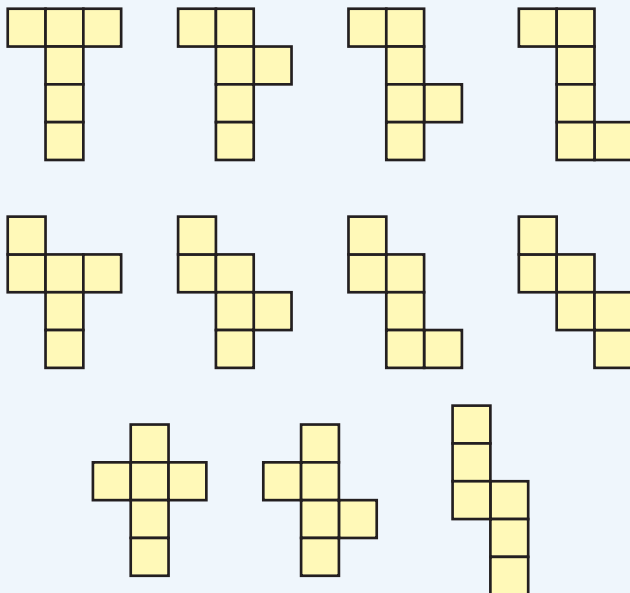
b)  $100 \times 100 = 10,000$  cubes in a layer

c)  $10,000 \times 100 = 1,000,000$  cubes in the box

$1,000,000\text{ cm}^3 = 1\text{ m}^3$

### Investigation 6:

A: There are 11 to choose from:



B: n and s

## Section D Learning Experiences:

### Assignment 1:

Shape 1:  $\frac{14}{2} + 4 - 1 = 10$        $7 + 4 - 1 = 10$       Correct

Shape 2:  $\frac{17}{2} + 6 - 1 = 13.5$        $8.5 + 6 - 1 = 13.5$       Correct

- An alternative is to count boxes and estimate, where each box is 1 square unit. This method works for both shapes.
- A second alternative is to break up the compound shapes into squares and rectangles:

Shape 1:  $4 \times 2 = 8$   
 $\frac{1}{2}(4 \times 1) = 2$        $8 + 2 = 10$

Shape 2:  $5 \times 2 = 10$   
 $3 \times 1 = 3$   
 $\frac{1}{2}(1 \times 1) = 0.5$        $10 + 3 + 0.5 = 13.5$

- Students should now be able to make a large poster version, showing the three methods to find the area.

### Assignment 2:

Formula for area of a triangle is  $\frac{1}{2}(\text{base}) \times (\text{height})$

For each of these triangles the base is staying fixed, and the perpendicular height is staying constant. Therefore, all the triangles will have the same area.

### Problem 1:

- Each strip has an area of  $1 \times 10 = 10 \text{ cm}^2$
- There are 4 strips  $4 \times 10 = 40 \text{ cm}^2$
- There are 4 instances when the strips overlap in  $1 \text{ cm}^2$  areas.

Therefore, the area covered  $= 40 - 4 = 36 \text{ cm}^2$

### Problem 2:

- $a + b = 15 \text{ cm}$
- $2b = 3a$
- $\therefore b = \frac{3}{2}a$  ( $1.5a$ )
- $\therefore a + 1.5a = 15$
- $\therefore 2.5a = 15$        $a = \frac{15}{2.5} = 6 \text{ cm}$
- $\therefore b = 9$

The total area  $= 15 \times 18 = 270 \text{ cm}^2$

**Section E Chapter review:**

1. The perimeter of a rectangle can be described by “twice the sum of the length and the **width**” of the rectangle.
2. The distance around a shape is called the **perimeter** of a shape.
3. Units of length are **millimetres**, centimetres, **metres** and kilometres.
4. The amount of surface covered by a shape is the **area** of the shape.
5. The area of any triangle can be described as “half the area of a rectangle with the same perpendicular height standing on the same **base**.”
6. To calculate the area of a triangle the **perpendicular** height of the triangle must be used.
7. To find the length of a side of a square we get the **square root** of the area.
8. Area is measured in cm **squared**.
9. Volume is measured in cm **cubed**.
10. A **net** of a solid can be used to calculate its surface area.
11. A rectangular solid has **6** sides (faces), **12** edges and **8** vertices.

# Unit of Learning:

## 10. Geometry 1

### Section A Unpacking the Learning Outcomes: \_\_\_\_\_

**GT.3:** Students should be able to investigate the concept of proof through their engagement with geometry so that they can:

a) perform constructions

|                      |                                                                                                                                                                                                                                                                                          |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>• what the words parallel and perpendicular mean</li> </ul>                                                                                                                                                                 |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>• draw: <ul style="list-style-type: none"> <li>– an angle of any given degree</li> <li>– a line parallel to a given line</li> <li>– a line perpendicular to a given line through a given point</li> </ul> </li> </ul> |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>• how to draw an angle greater than 180 degrees by calculating the remaining angle and using a protractor to draw this angle</li> </ul>                                                                               |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>• how important the skill is to accurately draw lines and angles</li> <li>• where these skills are used, such as engineering and architecture</li> </ul>                                                              |

b) recall and use the concepts, axioms, theorems, corollaries and converses, specified in *Geometry for Post-Primary School Mathematics*

|                      |                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>• how to name lines, line segments and rays</li> <li>• how to name different types of angles</li> <li>• how to name an angle using both a single letter, or 3 letters, and understand when it is appropriate to do either</li> </ul>                                                                                     |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>• use a protractor to measure an angle</li> <li>• recognise the geometry of pairs of angles and work out missing angles using one or more of these pairs</li> </ul>                                                                                                                                                |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>• the meaning of the following words in relation to pairs of angles: <ul style="list-style-type: none"> <li>– Supplementary</li> <li>– Vertically opposite</li> <li>– Alternate</li> <li>– Corresponding</li> <li>– Interior</li> <li>– Transversal</li> </ul> </li> <li>• the idea of collinear points</li> </ul> |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>• the importance of the study of Geometry, and it's place in the History of Mathematics</li> <li>• the importance of the study of Geometry in design and construction</li> </ul>                                                                                                                                   |

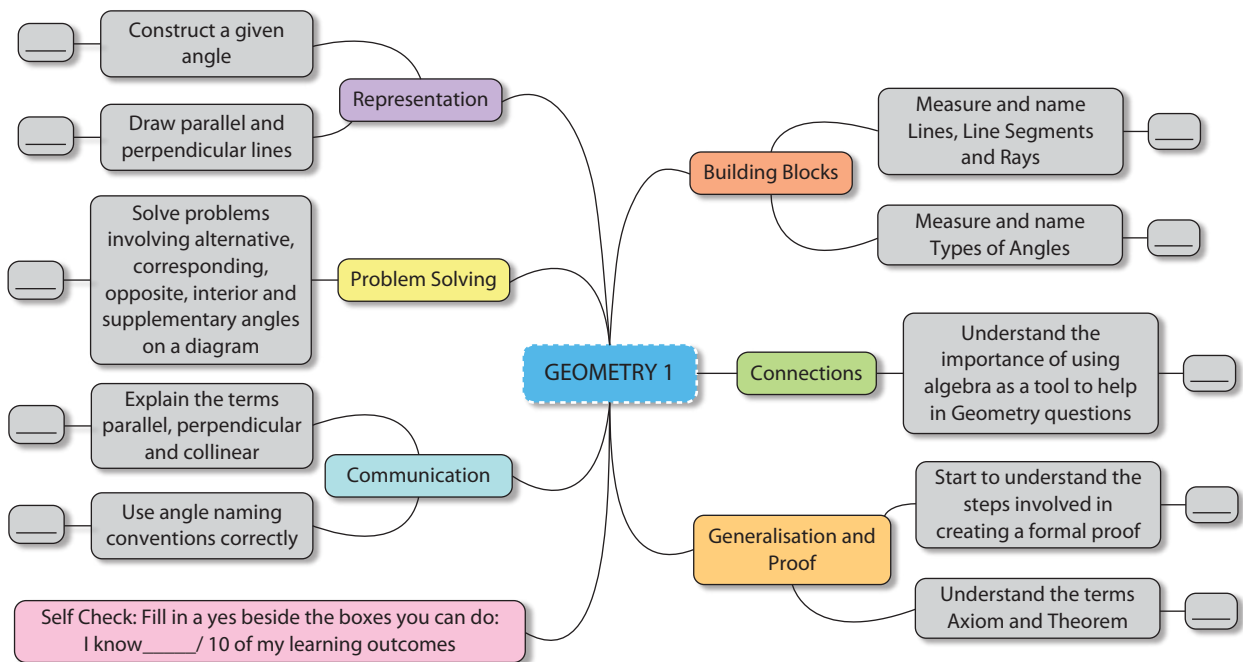


- c) display an understanding of the proofs of theorems

|                      |                                                                                                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>that the angles on a straight line add up to 180 degrees</li> </ul>                                     |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>verbally describe why different pairs of angles are equal and display them on diagrams</li> </ul> |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>how to create a simple proof for vertically opposite angles and interior angles</li> </ul>        |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>the important role that proofs of theorems play in Mathematics</li> </ul>                         |

## Section B Connecting the Unifying Strand:

All of these Learning Outcomes are summarised through the Lens of the Unifying Strand in the Mind Map, to help guide the Learning Experiences of the students:



## The Main Learning Outcomes from the Unifying Strand featured in this Unit of Learning are:

Students should be able to:

- U.1** recall and demonstrate understanding of the fundamental concepts and procedures that underpin each strand
- U.2** apply the procedures associated with each strand accurately, effectively, and appropriately
- U.5** make connections within and between strands
- U.6** make connections between mathematics and the real world
- U.7** make sense of a given problem, and if necessary mathematise a situation

- U.8** apply their knowledge and skills to solve a problem, including decomposing it into manageable parts and/or simplifying it using appropriate assumptions
- U.9** interpret their solution to a problem in terms of the original question
- U.11** generate general mathematical statements or conjectures based on specific instances
- U.12** generate and evaluate mathematical arguments and proofs
- U.13** communicate mathematics effectively: justify their reasoning, interpret their results, explain their conclusions, and use the language and notation of mathematics to express mathematical ideas precisely

## Section C Investigations:

### Investigation 1:

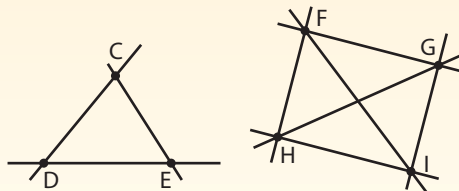
- $\angle C + \angle D = 180^\circ$  because they are on a straight line
- $\angle A + \angle D = 180^\circ$  because they are on a straight line
- $\angle A = \angle C$  as they both combine with  $\angle D$  to equal  $180^\circ$
- Vertically opposite angles are equal in measure. Similarly,  $\angle B = \angle D$ .

### Investigation 2:

- $A = C$  because they are corresponding angles
- $B + C = 180^\circ$  because they are on a straight line
- Therefore,  $A + B = 180^\circ$
- Therefore, interior angles add up to  $180^\circ$ .

## Section D Learning Experiences:

### Assignment 1:



| No. of Points | No. of Lines |
|---------------|--------------|
| 2             | 1            |
| 3             | 3            |
| 4             | 6            |
| 5             | 10           |

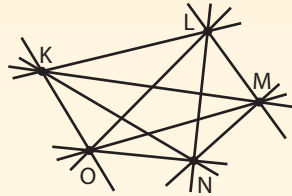
The total number of lines add up using this pattern:

2 points: 1

3 Points:  $2 + 1 = 3$

4 Points:  $3 + 2 + 1 = 6$

5 Points:  $4 + 3 + 2 + 1 = 10$



### Extension:

How many lines can be drawn through 6, 7 etc points:

6 Points:  $5 + 4 + 3 + 2 + 1 = 15$

7 Points:  $6 + 5 + 4 + 3 + 2 + 1 = 21$

### Assignment 2:

The poster should be colour coded by looking for:

- Alternate angles
- Corresponding angles
- Supplementary angles
- Vertically opposite

### Problem 1:

Angle BAC = 34 degrees

Angle ADC = 90 degrees

Angle DAC = 56 degrees

Angle ABC = 90 degrees

Angle BCA = 56 degrees

Angle ACD = 34 degrees

given

It is stated that it's a rectangle

$$90 - 34 = 56$$

It is stated that it's a rectangle

alternate angles

alternate angles

### Problem 2:

#### Discussion Points:

- How many angles do you have to measure with your protractor in order to know the size of all of them? (answer: 1)
- How could inaccuracies occur?
- Errors reading a protractor etc.

**Section E Chapter review:**

1. A line beginning at a point A and ending at a point B is called the **line - segment** [AB].
2. Three points on a straight line are said to be **collinear**.
3. An angle is a measure of **rotation**.
4. A full **circle** is also called a complete **rotation** and equals **360** degrees.
5. A  $90^\circ$  angle is called a **right** angle.
6.  **$180^\circ$**  is the same as half a rotation.
7. An **acute** angle is less than  $90^\circ$ .
8. A reflex angle is more than  **$180^\circ$** .
9. Angles on a **straight line** add up to  $180^\circ$ .
10. Vertically **opposite** angles are equal in measure.
11. A **protractor** is used to measure angles.
12. Parallel lines are always the **same distance** apart.
13. A line crossing parallel lines is called a **transversal**.
14. A line crossing parallel lines makes alternate and **interior** angles.
15. To draw a line parallel to a given line you need a **set square** and a **ruler**.
16. If  $C \parallel D$  then the lines C and D are **parallel**.
17. If  $E \perp F$  then the lines E and F are **perpendicular**.

# Unit of Learning:

## 11. Ratio and Proportion

### Section A Unpacking the Learning Outcomes:

**N.2** Students should be able to investigate equivalent representations of rational numbers so that they can:

**b)** use and understand ratio and proportion

|                      |                                                                                                                                                                                                                                                                                      |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>that when numbers are written in the form <math>a:b</math> they represent a ratio</li> <li>that proportions can be written as fractions, decimals or percentages</li> </ul>                                             |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>simplify a ratio by dividing by a common factor</li> <li>convert a ratio to the same units</li> <li>divide a given number into a given ratio</li> <li>find a total quantity when given part of a ratio</li> </ul> |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>the difference between ratio and proportion</li> <li>how to work with fraction ratios by converting them into whole numbers</li> </ul>                                                                            |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>the importance and significance of ratios and proportion in the world around them</li> <li>examples should as diluted drinks etc, where ratios are used in real life</li> </ul>                                   |

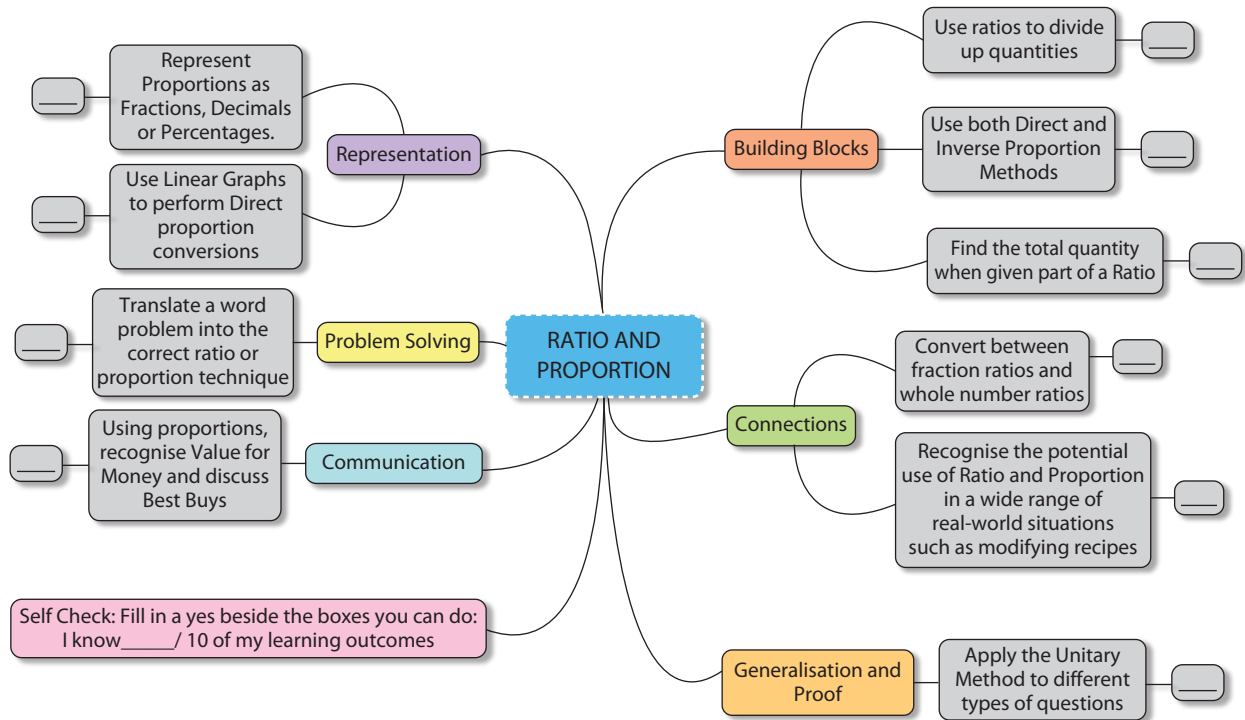
**N.3** Students should be able to investigate situations involving proportionality so that they can:

**b)** solve problems involving proportionality

|                      |                                                                                                                                                                                                                                                                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>that a direct proportion relationship will produce a straight-line graph through the origin</li> <li>how to decide which item is the best value when presented with different sizes or quantities</li> </ul>                                                  |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>apply the unitary method in order to solve problems involving proportion</li> <li>use conversion graphs to change between quantities that are both shown on the graph</li> <li>use proportionality to find quantities not shown on the graph</li> </ul> |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>the difference between direct and inverse proportion</li> <li>how to convert a word problem into the correct type of proportion problem</li> </ul>                                                                                                      |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>the connection between direct proportion and graphs</li> <li>the connection that graphs cutting the y axis, above or below the origin, involve some sort of fixed unit</li> </ul>                                                                       |

## Section B Connecting the Unifying Strand:

All of these Learning Outcomes are summarised through the Lens of the Unifying Strand in the Mind Map, to help guide the Learning Experiences of the students:



## The Main Learning Outcomes from the Unifying Strand featured in this Unit of Learning are:

Students should be able to:

- U.1** recall and demonstrate understanding of the fundamental concepts and procedures that underpin each strand
- U.2** apply the procedures associated with each strand accurately, effectively, and appropriately
- U.4** represent a mathematical situation in a variety of different ways, including: numerically, algebraically, graphically, physically, in words; and to interpret, analyse, and compare such representations
- U.5** make connections within and between strands
- U.6** make connections between mathematics and the real world
- U.7** make sense of a given problem, and if necessary mathematise a situation
- U.8** apply their knowledge and skills to solve a problem, including decomposing it into manageable parts and/or simplifying it using appropriate assumptions
- U.9** interpret their solution to a problem in terms of the original question
- U.13** communicate mathematics effectively: justify their reasoning, interpret their results, explain their conclusions, and use the language and notation of mathematics to express mathematical ideas precisely

## Section C Investigations:

### Investigation 1:

The easiest way to calculate the constant of Proportionality is to divide any y value by its corresponding x value. (if you want to convert from y to x)

#### Example:

$$\frac{2}{4.4} = 0.45$$

$$\frac{4}{8.8} = 0.45$$

$$\frac{6}{13.2} = 0.45 \quad \text{etc.}$$

This constant allows us to say “to convert from kg to pounds **multiple** by 0.45”

- i) 0.45
- ii)  $\frac{4.4}{2} = 2.2$
- iii)  $\frac{1.1}{5} = 0.22$
- iv)  $\frac{5}{1.1} = 4.55$
- v)  $\frac{10}{16} = 0.63$
- vi)  $\frac{16}{10} = 1.6$

### Investigation 2:

Paper dimensions in mm:

|           |           |
|-----------|-----------|
| <b>A3</b> | 297 × 420 |
| <b>A4</b> | 210 × 297 |
| <b>A5</b> | 148 × 210 |

- i) Area:  
A3: 124,740 mm<sup>2</sup>  
A4: 62,370 mm<sup>2</sup>  
A5: 31,080 mm<sup>2</sup>

As the sheets get smaller, the area is half that of the sheet before it. They have a ratio of 2:1

- ii)  $\frac{420}{297} = 1.41$   
 $\frac{297}{210} = 1.41$   
 $\frac{210}{148} = 1.41$

The ratio of each sheet's length to width is 1.41

(This number is often referred to as the Golden Ratio. It appears in nature all around us)

#### Extension:

Get students to do a research project on the Golden Ratio.

## Section D Learning Experiences:

### Assignment 1:

#### Discussion Points:

- how best to collect the data?
- which is easier to understand, ratio or proportion?
- are the different categories giving similar ratios? Why not?

### Assignment 2:

#### Points to note:

- You may need to estimate when trying to get all items into the same units
- You can group very small ingredients together

### Problem 1:

| Men | Days | Fields | Types of proportion |
|-----|------|--------|---------------------|
| 5   | 5    | 5      |                     |
| 1   | 25   | 5      | Direct              |
| 1   | 5    | 1      | Inverse             |

### Problem 2:

- Each of the 5 tyres is not used for an equal part of the journey
- We must divide the total journey by 5 to get the time not being used.  $\frac{30,000}{5} = 6,000$  km
- Therefore, each tyre is used for  $30,000 - 6,000 = 24,000$  km

OR

- Total distance travelled by four wheels =  $4 \times 30,000 = 120,000$  km
- Shared equally between 5 tyres =  $120,000 \div 5 = 24,000$  km



**Section E Chapter review:**

1. Ratios compare **part** to **part** of the quantity.
2. Proportions compare **part** to **whole** of any quantity.
3. Ratios can be simplified by dividing by a **common factor**.
4. The same **units** must be used when comparing two quantities using ratios.
5. When dividing any quantity in ratio, first find the total number of **parts** needed.
6. It is easier to compare proportions when they are given as **decimals**.
7. To change from one quantity (10 kg) of a substance to a different quantity (3 kg) of a substance, we use the **unitary method**, i.e. we find **1** kg first.
8. A direct proportion when plotted produces a **straight-line** graph through the **origin**.
9. We deal with two types of proportion, direct and **inverse** proportion.

# Unit of Learning:

## 12. Statistics

### Section A Unpacking the Learning Outcomes:

**SP.3** Students should be able to carry out a statistical investigation which includes the ability to:

- generate a statistical question
- plan and implement a method to generate and/or source unbiased, representative data, and present this data in a frequency table

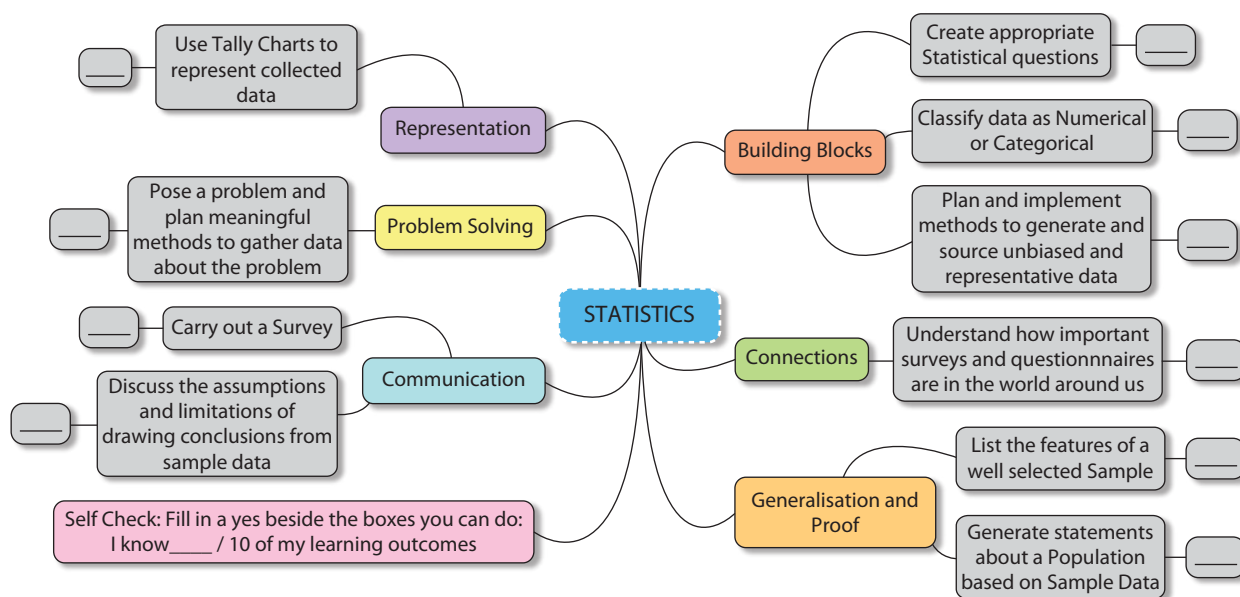
|                      |                                                                                                                                                                                                                                                                                                                             |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | <p>Students should know:</p> <ul style="list-style-type: none"> <li>The characteristics of a good statistical question i.e. <ul style="list-style-type: none"> <li>clear and short</li> <li>not leading</li> <li>not personal</li> <li>give options</li> </ul> </li> <li>the steps involved in planning a survey</li> </ul> |
| <b>Skills</b>        | <p>Students should be able to:</p> <ul style="list-style-type: none"> <li>complete a tally chart</li> <li>select an appropriate sample for their statistical question</li> <li>select a sample that is representative of the population</li> <li>select a sample that is not biased</li> </ul>                              |
| <b>Understanding</b> | <p>Students should understand:</p> <ul style="list-style-type: none"> <li>the data handling cycle</li> <li>the importance of each step in a statistical investigation</li> <li>how to read data when presented in a frequency table</li> </ul>                                                                              |
| <b>Value</b>         | <p>Students should:</p> <ul style="list-style-type: none"> <li>appreciate the different methods of collecting data, from surveys and questionnaires, to experiments and research</li> <li>be able to choose the correct method to generate any requested data</li> </ul>                                                    |

- classify data (categorical, numerical)
- discuss the assumptions and limitations of conclusions drawn from sample data

|                      |                                                                                                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | <p>Students should know:</p> <ul style="list-style-type: none"> <li>how to identify misleading data</li> <li>about primary and secondary data</li> <li>what a census is</li> </ul>                                                |
| <b>Skills</b>        | <p>Students should be able to:</p> <ul style="list-style-type: none"> <li>classify any given data into either numerical or categorical</li> <li>select an appropriate sample size for any given population</li> </ul>             |
| <b>Understanding</b> | <p>Students should understand:</p> <ul style="list-style-type: none"> <li>the pros and cons of using both a large or small sample size</li> <li>why samples are used instead of collecting data from whole populations</li> </ul> |
| <b>Value</b>         | <p>Students should appreciate:</p> <ul style="list-style-type: none"> <li>that not all statistics they are presented with in real life are as meaningful as they might seem</li> </ul>                                            |

## Section B Connecting the Unifying Strand:

All of these Learning Outcomes are summarised through the Lens of the Unifying Strand in the Mind Map, to help guide the Learning Experiences of the students:



## The Main Learning Outcomes from the Unifying Strand featured in this Unit of Learning are:

Students should be able to:

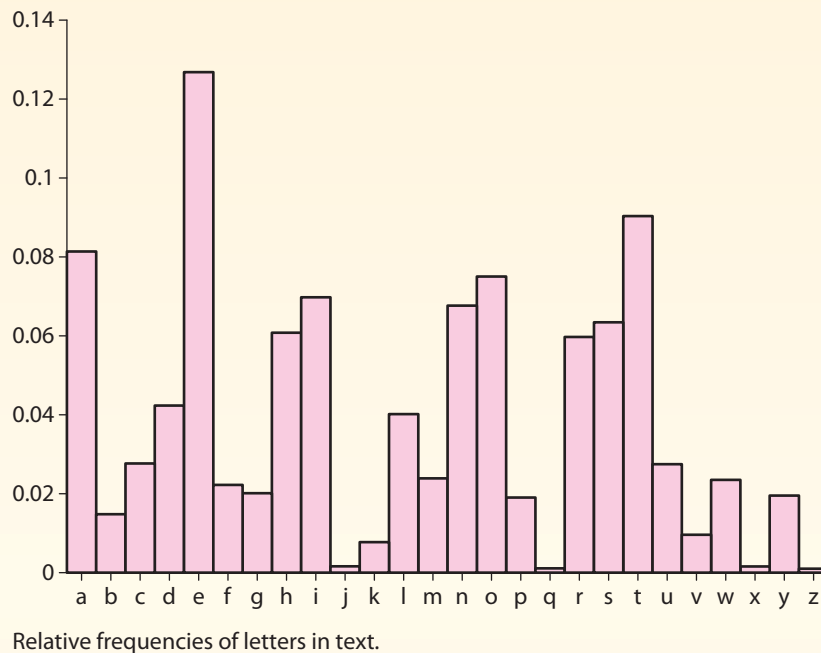
- U.1** recall and demonstrate understanding of the fundamental concepts and procedures that underpin each strand
- U.2** apply the procedures associated with each strand accurately, effectively, and appropriately
- U.4** represent a mathematical situation in a variety of different ways, including: numerically, algebraically, graphically, physically, in words; and to interpret, analyse, and compare such representations
- U.5** make connections within and between strands
- U.6** make connections between mathematics and the real world
- U.7** make sense of a given problem, and if necessary mathematise a situation
- U.8** apply their knowledge and skills to solve a problem, including decomposing it into manageable parts and/or simplifying it using appropriate assumptions
- U.9** interpret their solution to a problem in terms of the original question
- U.13** communicate mathematics effectively: justify their reasoning, interpret their results, explain their conclusions, and use the language and notation of mathematics to express mathematical ideas precisely

## Section C Learning Experiences:

### Assignment 1:

#### Discussion Points:

- Students should notice when they compare their results the numbers for each vowel are similar to other students work.
- This chart shows the relative frequency of all of the 26 letters of the English alphabet:



Does this match their findings in relation to the vowels?

#### Extension:

- Do students think that all languages will have the same distribution?

### Assignment 2:

RANDOM: made, done, or happening without method or conscious decision.

Random Sample from the class:

- Names out of a hat
- Random number generator
- First 5 students to walk into the classroom
- Etc.

Random Sample from School:

- Names from a hat
- Assign numbers and select
- Use roll numbers and select
- First 5 students to arrive at school
- Etc.

**Extension:****Discussion Points**

- What is the most common number of children in a family? (modal)
- Did the class percentages match the sample percentage? Why/Why not?
- Is 5 a good sample size?
- If your population had (i) 100 (ii) 1000 people in it, what do you think a good sample size would be?

| Sample size<br>( <i>N</i> ) | Margin of error<br>(fraction) | Margin of error<br>(percentage) |
|-----------------------------|-------------------------------|---------------------------------|
| 10                          | 0.316                         | 31.6                            |
| 20                          | 0.224                         | 22.4                            |
| 50                          | 0.141                         | 14.1                            |
| 100                         | 0.100                         | 10.0                            |
| 200                         | 0.071                         | 7.1                             |
| 500                         | 0.045                         | 4.5                             |
| 1000                        | 0.032                         | 3.2                             |
| 2000                        | 0.022                         | 2.2                             |
| 5000                        | 0.014                         | 1.4                             |
| 10000                       | 0.010                         | 1.0                             |

You can quickly see from the table that results from a survey with only 10 random participants are not reliable. The margin of error in this case is roughly 32%. This means that if you found, for example, that 6 out of your 10 participants (60%) had a fear of heights, then the actual proportion of the population with a fear of heights could vary by  $\pm 32\%$ . In other words, the actual proportion could be as low as 28% ( $60 - 32$ ) and as high as 92% ( $60 + 32$ ). With a range that large, your small survey isn't saying much.

If you increase the sample size to 100 people, your margin of error falls to 10%. Now if 60% of the participants reported a fear of heights, there would be a 95% probability that between 50 and 70% of the total population have a fear of heights. Now you're getting somewhere. If you want to narrow the margin of error to  $\pm 5\%$ , you have to survey 500 randomly-selected participants. The bottom line is, you need to survey a *lot* of people before you can start having any confidence in your results.

**Extension:**

- Do you think your class answers would be the same as those in: (i) another part of Ireland (ii) another country?

**Assignment 3:****Discussion Points:**

- Were you allowed pick your own groups, or did your teacher put you in groups?
- Will this impact on your group answers?
- Do you think your group is a representative sample for your class/year/school?

*Assignment 4:*

| Total number of students | Main method of travel | Number of males | Percentage of males | Number of females | Percentage of females |
|--------------------------|-----------------------|-----------------|---------------------|-------------------|-----------------------|
| 108,400                  | Bus                   | 52,639          | 35.8%               | 55,761            | 37.5%                 |
| 101,036                  | Car                   | 48,418          | 32.9%               | 52,618            | 35.4%                 |
| 74,346                   | Walked                | 36,416          | 24.8%               | 37,930            | 25.4%                 |
| 7,377                    | Cycled                | 6,856           | 4.7%                | 521               | 0.4%                  |
| 4672                     | Train                 | 2,673           | 1.8%                | 1,999             | 1.3%                  |
|                          |                       |                 |                     |                   |                       |
| 295,831                  |                       | 147,002         |                     | 148,829           |                       |

**Discussion points:**

- Rural / Urban School
- School Bus option
- Location of nearest train station
- Differences/similarities of male/female modes of transport (look at the cycle option)

**Section D Chapter review:**

1. To answer statistical questions you need to do a **survey**.
2. **Data** is information that you collect.
3. A tally sheet usually counts in groups of **five**.
4. Data in the form of numbers is called **numerical** data.
5. Types of houses is an example of **categorical** data.
6. The duration of a basketball match is an example of **numerical** data.
7. Everybody who could be included in a survey is called the **population**.
8. When sampling it is very important to avoid **bias**.
9. A **random** sample is one in which every member of the population has the same chance of being selected.
10. In questionnaires avoid **leading** questions because they suggest a particular response.
11. In questionnaires avoid **personal** questions because they may cause upset or embarrassment.

# Unit of Learning:

## 13. Coordinates

### Section A Unpacking the Learning Outcomes:

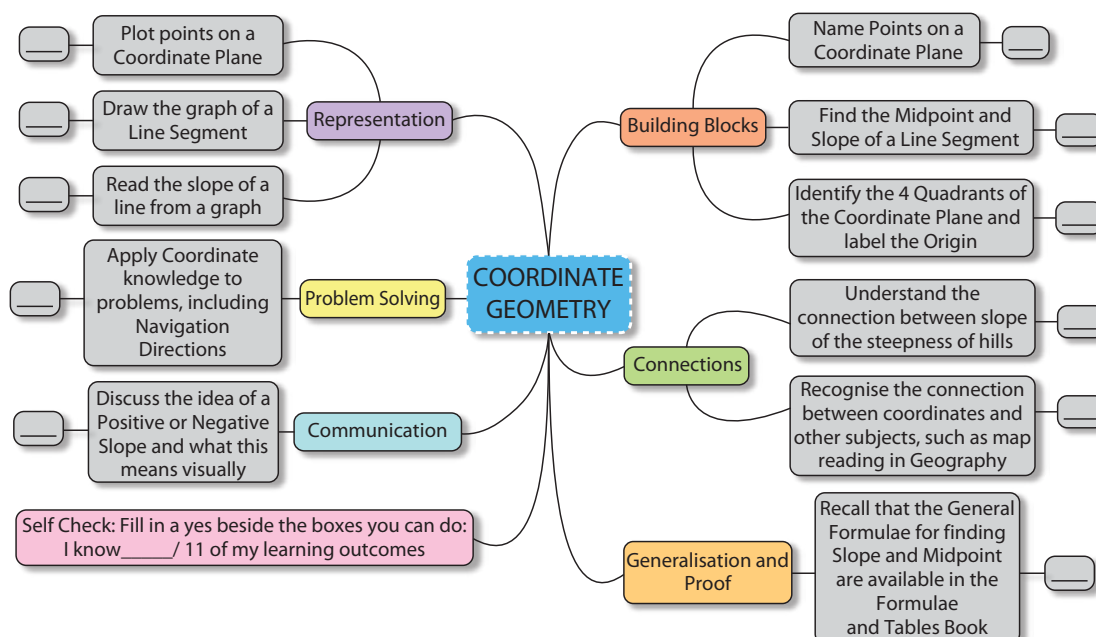
**GT.5** Students should be able to investigate properties of points, lines and line segments in the co-ordinate plane so that they can:

a) find and interpret midpoint and slope

|                      |                                                                                                                                                                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>• how to recognise a Coordinated Plane</li> <li>• what and where the origin is</li> <li>• how to read the slope of a line segment from a graph using Rise/Run</li> </ul>                |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>• draw a grid and label it correctly</li> <li>• name the 4 quadrants</li> <li>• estimate the midpoint of a line segment from a graph</li> </ul>                                   |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>• how to use the midpoint formula</li> <li>• how to use the slope formula</li> <li>• that they can sketch the graph of a line, given the slope and a point on the line</li> </ul> |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>• the importance of coordinates in the world around them e.g. map reading in Geography, Satellite Navigation Systems in cars</li> </ul>                                           |

### Section B Connecting the Unifying Strand:

All of these Learning Outcomes are summarised through the Lens of the Unifying Strand in the Mind Map, to help guide the Learning Experiences of the students:



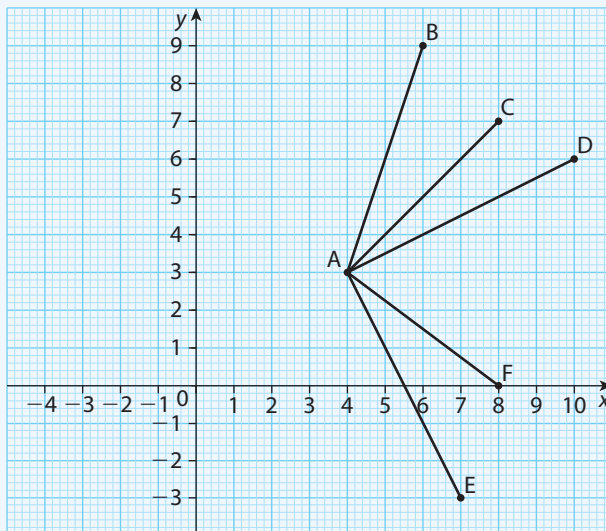
## The Main Learning Outcomes from the Unifying Strand featured in this Unit of Learning are:

Students should be able to:

- U.1** recall and demonstrate understanding of the fundamental concepts and procedures that underpin each strand
- U.2** apply the procedures associated with each strand accurately, effectively, and appropriately
- U.4** represent a mathematical situation in a variety of different ways, including: numerically, algebraically, graphically, physically, in words; and to interpret, analyse, and compare such representations
- U.5** make connections within and between strands
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- U.9** interpret their solution to a problem in terms of the original question
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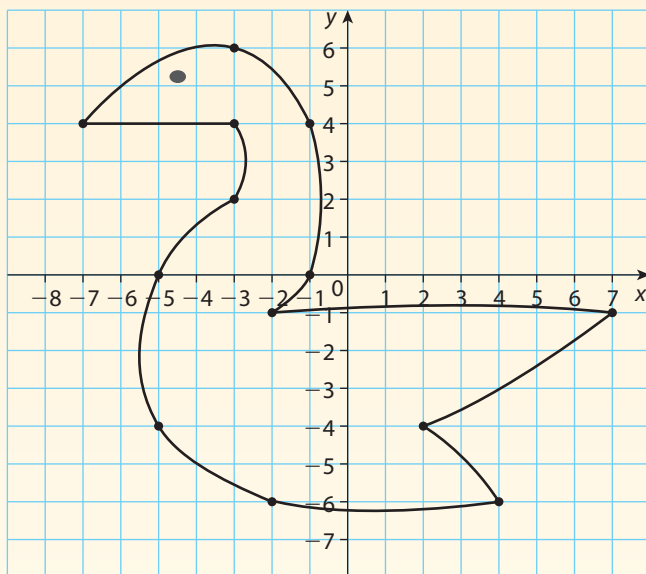
## Section C Investigations:

### *Investigation 1:*





## Section D Learning Experiences:

*Assignment:**Assignment 1:*

- a) A(2, 3)  
B(12, 3)  
C(12, 9)  
D(12, 7)  
E(12, 5)
- b) [AC] slope =  $\frac{6}{10} = 0.6$   
[AD] slope =  $\frac{4}{10} = 0.4$   
[AE] slope =  $\frac{2}{10} = 0.2$   
[AB] slope =  $\frac{0}{10} = 0$
- c) As the line segment is lowered, the slope gets closer and closer to zero, and will equal zero when horizontal.
- d) A horizontal line has a slope of zero.
- e) H(1, 1)  
J(3, 11)  
K(2, 11)  
L(1.4, 11)  
M(1.2, 11)  
N(1.1, 11)  
I(1, 11)

f) [HJ] slope =  $\frac{10}{2} = 5$

[HK] slope =  $\frac{10}{1} = 10$

[HL] slope =  $\frac{10}{0.4} = 25$

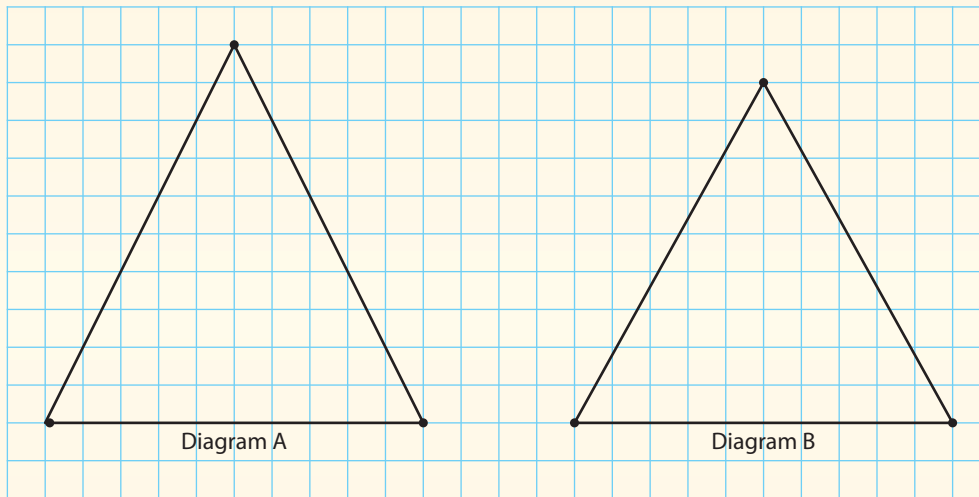
[HM] slope =  $\frac{10}{0.2} = 50$

[HN] slope =  $\frac{10}{0.1} = 100$

[HI] slope =  $\frac{10}{0} = \text{UNDEFINED}$

- g) As the move closer to I the slope increases rapidly. Once it hits vertical, this means there is no RUN, so we are dividing by zero. In mathematics, this is undefined.

### Assignment 2:



Most students will make the mistake of drawing the diagram like A. The method here is count the boxes across the bottom (10) and then place the top vertex 10 boxes up, above the midpoint of the base. This will result in the 2 side lengths being longer than the base.

To do this accurately, students would need to do an equilateral triangle construction, using a compass. However, if students measure the base with a ruler, they may be able to create a fairly accurate version, by trial and error, like in Diagram B.

### Problem 1:

Assumption 1: The line [AF] is perpendicular to [BC]

$$\text{Area of a triangle} = \frac{1}{2}(\text{base})(\text{height})$$

$$\text{Base} = 6 \quad (10 - 4)$$

We assume F is at the midpoint of [BC] = (7, 4)

Therefore, the height is 4 (8 - 4)

$$\text{Area of a triangle} = \frac{1}{2}(6)(4) = 12 \text{ square units}$$

**Section E Chapter review:** \_\_\_\_\_

1. To fix a point on a plane we need its **coordinates**.
2. The general symbol for a point on the plane is **( $x$ ,  $y$ )**.
3. The point (0, 0) is called the **origin**.
4. The coordinated plane is formed by  **$x$**  and  **$y$**  axes, drawn at right angles to one another.
5. Each plane is divided into 4 **quadrants** by the  $x$  and  $y$  **axes**.
6. To find the mid-point of a line segment add the  $x$ -values and divide by **2**, add the  $y$ -values and divide by **2**.
7. The slope of a line measures the **steepness** of the line.
8. The slope of the line is calculated by dividing the **rise** by the **run**.
9. The symbol for the slope of a line is  **$m$** .
10. A **horizontal** line has a slope of 0.

# Unit of Learning:

## 14. Solving equations

### Section A Unpacking the Learning Outcomes: \_\_\_\_\_

**AF.2:** Students should be able to investigate situations in which letters stand for quantities that are variable so that they can:

- generate and interpret expressions in which letters stand for numbers
- find the value of expressions given the value of the variables

|                      |                                                                                                                                                                                                                                                                     |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>how to represent unknown numbers using symbols or letters</li> <li>that when an expression contains more than one operation, we follow BIRDMAS</li> </ul>                                              |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>find the value of an expression by substituting in given numbers for letters</li> <li>translate an expression into an English statement and vice versa</li> </ul>                                |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>the meaning of the words variable, term, expression, like/unlike terms, constant and coefficient</li> </ul>                                                                                      |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>the use of letters in formulae they would have already come across in primary school to represent area, volume etc. and appreciate the connections between these formulae and algebra</li> </ul> |

- use the concept of equality to generate and interpret equations

|                      |                                                                                                                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>the mathematical meaning of English words that appear in problems</li> </ul>                                                     |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>use the balancing method to solve an equation</li> <li>translate an equation into a word problem and vice versa</li> </ul> |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>that they need to use the skills they learnt in other areas of algebra in order to simplify linear equations</li> </ul>    |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>that being able to solve an equation is a fundamental mathematical skill that will be used in all other strands</li> </ul> |

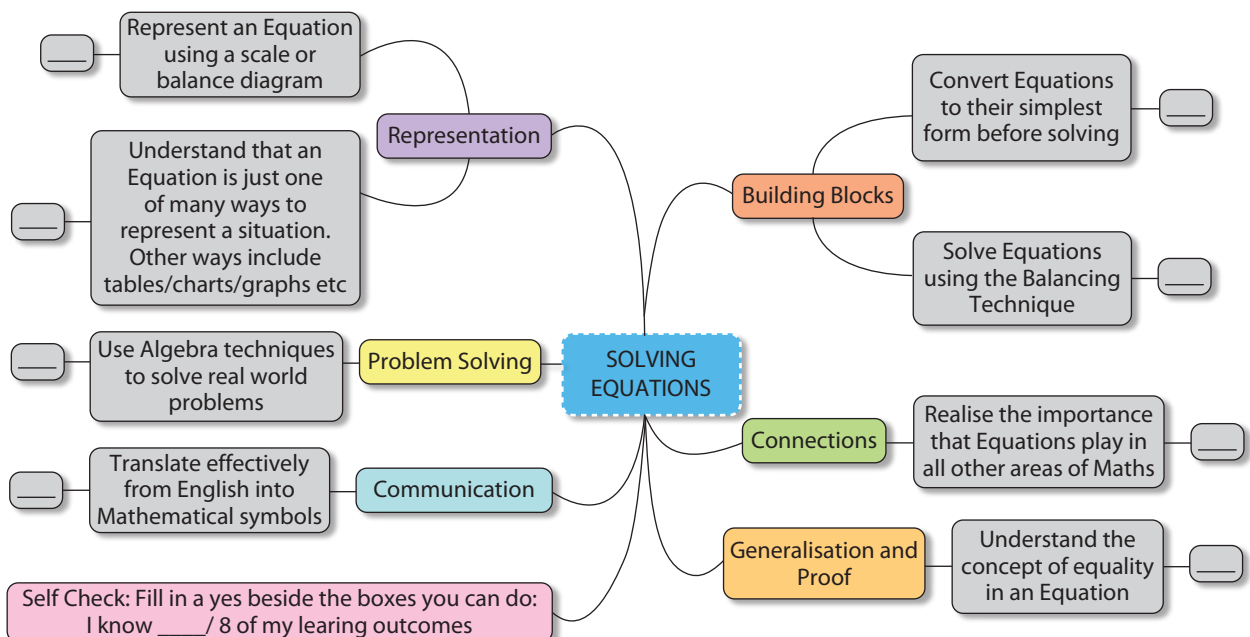
**AF.3:** Students should be able to apply the properties of arithmetic operations to generate equivalent expressions so that they can develop and use appropriate strategies to:

- a) simplify
  - i) linear expressions in one or more variables with coefficients in  $\mathbb{Q}$
  - ii) quadratic expressions in one variable with coefficients in  $\mathbb{Z}$
  - iii) expressions of the form  $a / (bx + c)$ , where  $a, b, c \in \mathbb{Z}$
- b) multiply expressions of the form
  - i)  $a(bx + cy + d)$ ;  $a(bx^2 + cx + d)$  and  $ax(bx^2 + cx + d)$ , where  $a, b, c, d \in \mathbb{Z}$
  - ii)  $(ax + b)(cx + d)$  where  $a, b, c, d \in \mathbb{Z}$

|                      |                                                                                                                                                                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>like terms, recognising that <math>x^2</math> and <math>x</math> are not like terms</li> <li>that when an expression contains more than one operation, we follow BIRDMAS</li> </ul>                                                             |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>simplify an expression by grouping like terms together</li> <li>multiply single brackets by a constant or variable or both</li> <li>multiply 2 brackets, each with 2 terms, using an array model</li> </ul>                               |
| <b>Understanding</b> | Students should understand that: <ul style="list-style-type: none"> <li>the rules for multiplying or dividing integers apply to variables as well</li> <li>they can apply their knowledge of indices when multiplying or dividing variables with index numbers i.e. <math>(x^2)(x^3) = x^5</math></li> </ul> |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>that when we multiply or divide expressions a new version (usually simpler) of the original expressions is formed</li> <li>the connection between area and multiplication by examining array models</li> </ul>                            |

## Section B Connecting the Unifying Strand:

All of these Learning Outcomes are summarised through the Lens of the Unifying Strand in the Mind Map, to help guide the Learning Experiences of the students:



## The Main Learning Outcomes from the Unifying Strand featured in this Unit of Learning are:

Students should be able to:

- U.1** recall and demonstrate understanding of the fundamental concepts and procedures that underpin each strand
- U.2** apply the procedures associated with each strand accurately, effectively, and appropriately
- U.3** recognise that equality is a relationship in which two mathematical expressions have the same value
- U.4** represent a mathematical situation in a variety of different ways, including: numerically, graphically, physically and in words
- U.5** make connections within and between strands
- U.6** make connections between mathematics and the real world
- U.7** make sense of a given problem, and if necessary mathematise a situation
- U.8** apply their knowledge and skills to solve a problem, including decomposing it into manageable parts and/or simplifying it using appropriate assumptions
- U.9** interpret their solution to a problem in terms of the original question
- U.13** communicate mathematics effectively: justify their reasoning, interpret their results, explain their conclusions, and use the language and notation of mathematics to express mathematical ideas precisely

## Section C Investigations:

### *Investigation 1:*

Methods we hope to see students use:

- Counting
- Table
- List
- Graph
- Equation
- Diagram
- Etc.

i) €115               $25 + 6(15)$

ii) €265              $25 + 16(15)$

iii) €805             $25 + 52(15)$

iv) Equation – same one can be used over and over, just with different substitution

v) 12

vi)  $25 + 15x = 200$

$$\begin{array}{r} -25 \qquad -25 \\ \hline \end{array}$$

$$15x = 175$$

$$x = 11.666\dots$$

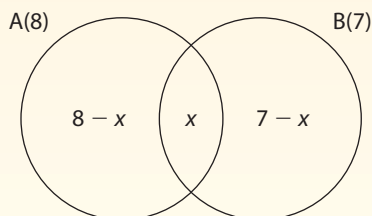
She needs to save for 12 weeks

*Investigation 2:***Examples:**

| +                                                | −                                                 | ×                                                                             | ÷                                                         | =                                                 |
|--------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------|
| sum<br>increased<br>combine<br>plus<br>more than | difference<br>decreased<br>less than<br>take away | product<br>times<br>factor<br>twice ( $2\times$ 's)<br>triple ( $3\times$ 's) | quotient<br>divide<br>into<br>out of<br>split<br>break up | is<br>total<br>result<br>same as<br>equivalent to |

**Section D Learning Experiences:***Assignment 1:*

- Place 3 coins on each side of the scales
- One side will be lighter, keep these 3 coins.
- Select 2 out of the 3 coins.
- Place one on each side.
- If the scales are balanced, the lighter one is the one not on the scales.
- If the scales are not balanced, you can clearly see which is the lighter one.

*Assignment 2:*

$$\begin{aligned}(8 - x) + x + (7 - x) &= 10 \\ 15 - x &= 10 \\ \Rightarrow x &= 5\end{aligned}$$

$$\begin{aligned}A \setminus B &= 8 - x = 8 - 5 = 3 \\ B \setminus A &= 7 - x = 7 - 5 = 2\end{aligned}$$

$$\Rightarrow 3 + 2 + 5 = 10$$

*Problem 1:*

$$R + Q + 8 = 16$$

$$R + Q = 8$$

$$R + Q + P + P = 14$$

$$8 + 2P = 14$$

$$2P = 6 \quad \Rightarrow P = 3$$

$$2P + 4 + R = 16$$

$$6 + 4 + R = 16 \quad \Rightarrow R = 6$$

$$4 + R + Q + P = 15$$

$$4 + 6 + Q + 3 = 15$$

$$13 + Q = 15 \quad \Rightarrow Q = 2$$

$$2Q + 2S = 14$$

$$4 + 2S = 14$$

$$2S = 10 \quad \Rightarrow S = 5$$

### Problem 2:

$$3(6 + x) = (40 + x)$$

$$18 + 3x = 40 + x$$

$$2x = 22$$

$$x = 11$$

In 11 years Michelle will be  $6 + 11 = 17$  and I will be  $40 + 11 = 51$ .

$$17 \times 3 = 51.$$

## Section E Chapter review:

1. In algebra we let  $x, y, a, b$ , etc. to represent an **unknown** quantity.
2. Information in problems is changed into **Mathematical symbols** using algebra.
3. Central to every equation is the **equal** sign.
4. If we find a value for  $x, y, a$ , etc., we have **solved** the equation.
5. When solving an equation, we use the "**Balancing**" method.



# Unit of Learning:

## 15. Geometry 2

### Section A Unpacking the Learning Outcomes: \_\_\_\_\_

**GT.3:** Students should be able to investigate the concept of proof through their engagement with geometry so that they can:

a) perform constructions

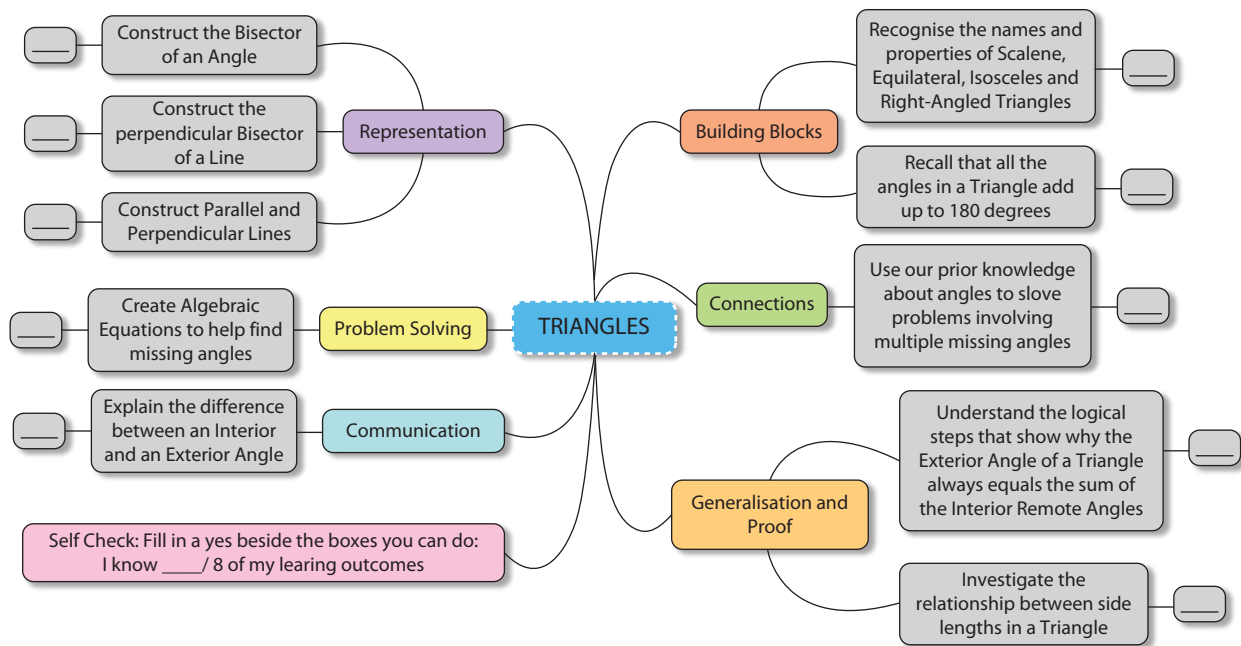
|                      |                                                                                                                                                                                                                                                                                                                                                  |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>• what the words parallel and perpendicular mean</li> <li>• how to construct:               <ul style="list-style-type: none"> <li>– the bisector of a given angle</li> <li>– the bisector of a line segment</li> <li>– a line divided into 3 equal sections</li> </ul> </li> </ul> |
| <b>Skills</b>        | Students should be able to draw: <ul style="list-style-type: none"> <li>• an angle of any given degree</li> <li>• a line parallel to a given line</li> <li>• a line perpendicular to a given line through a given point</li> </ul>                                                                                                               |
| <b>Understanding</b> | Students should understand how: <ul style="list-style-type: none"> <li>• to draw an angle greater than 180 degrees by calculating the remaining angle and using a protractor to draw this angle</li> </ul>                                                                                                                                       |
| <b>Value</b>         | Students should appreciate <ul style="list-style-type: none"> <li>• the importance of the skill to accurately draw and bisect lines and angles</li> <li>• places that these skills are used, such as engineering and architecture</li> </ul>                                                                                                     |

b) recall and use the concepts, axioms, theorems, corollaries and converses, specified in *Geometry for Post-Primary School Mathematics*

|                      |                                                                                                                                                                                                                                                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know: <ul style="list-style-type: none"> <li>• how to name a triangle and the notation that is involved</li> <li>• that the angles in a triangle add up to 180 degrees</li> <li>• that the interior opposite angles add to give the exterior angle</li> </ul>                                                       |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>• use a protractor to measure an angle</li> <li>• demonstrate that the angles in a triangle add up to 180° using an informal proof</li> </ul>                                                                                                                    |
| <b>Understanding</b> | Students should understand: <ul style="list-style-type: none"> <li>• the following words in relation to triangles:               <ul style="list-style-type: none"> <li>– Exterior Angle</li> <li>– Equilateral</li> <li>– Isosceles</li> <li>– Scalene</li> </ul> </li> <li>• how to find a missing angle in a triangle</li> </ul> |
| <b>Value</b>         | Students should appreciate: <ul style="list-style-type: none"> <li>• the importance of the study of Geometry, and its place in the History of Mathematics</li> </ul>                                                                                                                                                                |

## Section B Connecting the Unifying Strand:

All of these Learning Outcomes are summarised through the Lens of the Unifying Strand in the Mind Map, to help guide the Learning Experiences of the students:



### The Main Learning Outcomes from the Unifying Strand featured in this Unit of Learning are:

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- U.5** make connections within and between strands
- U.6** make connections between mathematics and the real world
- U.7** make sense of a given problem, and if necessary mathematise a situation
- U.8** apply their knowledge and skills to solve a problem, including decomposing it into manageable parts and/or simplifying it using appropriate assumptions
- U.9** interpret their solution to a problem in terms of the original question
- U.11** generate general mathematical statements or conjectures based on specific instances
- U.12** generate and evaluate mathematical arguments and proofs
- U.13** communicate mathematics effectively: justify their reasoning, interpret their results, explain their conclusions, and use the language and notation of mathematics to express mathematical ideas precisely

## Section C Investigations:

### Investigation 1:

Conclusion: The angles inside a triangle add up to the same as a straight line  
i.e. 180 degrees.

### Investigation 2:

| Side 1 | Side 2 | Side 3 | Yes/No |
|--------|--------|--------|--------|
| 5 cm   | 7 cm   | 10 cm  | Yes    |
| 4 cm   | 5 cm   | 11 cm  | No     |
| 5 cm   | 4 cm   | 3 cm   | Yes    |
| 12 cm  | 5 cm   | 5 cm   | No     |
| 6 cm   | 6 cm   | 6 cm   | Yes    |

Any two sides of a triangle must add up to more than the third side, otherwise a triangle cannot be formed.

## Section D Learning Experiences:

### Assignment 1:

|                        | Triangle | Square | Pentagon |
|------------------------|----------|--------|----------|
| Number of sides        | 3        | 4      | 5        |
| Number of angles       | 3        | 4      | 5        |
| Sum of all the angles  | 180      | 360    | 540      |
| The size of each angle | 60       | 90     | 108      |

#### Extension:

- Continue the table for a hexagon, heptagon etc
- Try to formulate a rule for the size of each angle within any regular sided shape

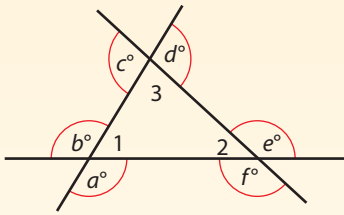
### Problem 1:

$$180 - 75 = 105$$

$$180 - 165 = 15$$

$$105 + 15 = 120$$

$$X^\circ = 180^\circ - 120^\circ = 60^\circ$$

**Problem 2:**

$$a = 3 + 2$$

$$b = 3 + 2$$

$$c = 1 + 2$$

$$d = 1 + 2$$

$$e = 1 + 3$$

$$f = 1 + 3 \quad +$$

$$a + b + c + d + e + f = 3 + 2 + 3 + 2 + 1 + 2 + 1 + 2 + 1 + 3 + 1 + 3$$

$$a + b + c + d + e + f = (1 + 2 + 3) + (1 + 2 + 3) + (1 + 2 + 3) + (1 + 2 + 3)$$

$$a + b + c + d + e + f = 180 + 180 + 180 + 180$$

$$a + b + c + d + e + f = 720 \text{ degrees}$$

**Or**

$$3 \times 360^\circ - 2(1 + 2 + 3) = 720^\circ$$

$$3 \times 360^\circ - 2(180^\circ) = 720^\circ$$

**Section E Chapter review:**

1. Each corner of a triangle is called a **vertex**.
2. The **interior angles** of a triangle add up to  $180^\circ$ .
3. When any side of a triangle is extended an **exterior** angle is formed.
4. The **exterior** angle of a triangle is equal in size to the sum of the interior remote angles.
5. All the angles of an **equilateral** triangle measure  $60^\circ$ .
6. In an isosceles triangle only **2** of the sides have the same length.
7. A triangle with all three sides the same length is called an **equilateral** triangle.
8. A scalene triangle has three **different** sides and three unequal **angles**.
9. In a right-angled triangle the side opposite the right angle is the **hypotenuse**.
10. In a right-angled triangle the longest side is the **hypotenuse**.
11. To construct a bisector of an angle a **compass** and **ruler** are needed.
12. To construct a line parallel to a given line a **ruler** and **set square** are needed.

# Unit of Learning:

## 16. Presenting Data

### Section A Unpacking the Learning Outcomes:

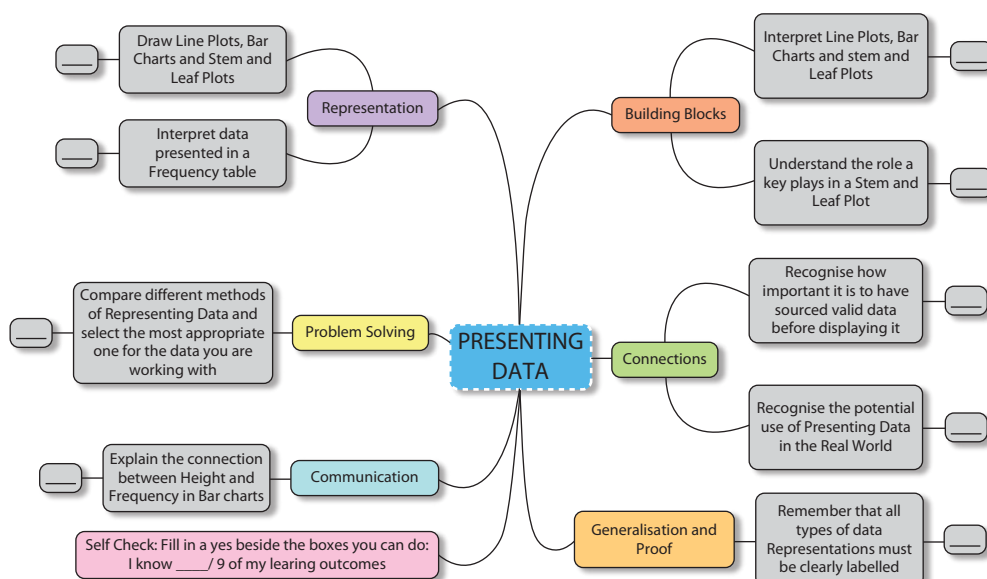
**SP.3** Students should be able to carry out a statistical investigation which includes the ability to:

- d) select, draw and interpret appropriate graphical displays of univariate data, including bar charts, line plots, ordered stem and leaf plots

|                      |                                                                                                                                                                                                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know how to: <ul style="list-style-type: none"> <li>interpret Line Plots, Bar Charts and Stem and Leaf Plots</li> <li>select the most appropriate chart for the data they wish to present</li> </ul>                                                 |
| <b>Skills</b>        | Students should be able to: <ul style="list-style-type: none"> <li>accurately draw: <ul style="list-style-type: none"> <li>Line Plots</li> <li>Bar Charts</li> <li>Stem and Leaf Plots</li> </ul> </li> <li>interpret data presented in a Frequency table</li> </ul> |
| <b>Understanding</b> | Students should understand the: <ul style="list-style-type: none"> <li>importance of the key in a Stem and Leaf Plot</li> <li>connection between Height and Frequency in Bar Charts</li> </ul>                                                                       |
| <b>Value</b>         | Students should: <ul style="list-style-type: none"> <li>appreciate how important it is to source valid and unbiased data before displaying it</li> <li>recognise charts and graphs they see around them, for example in advertising etc.</li> </ul>                  |

### Section B Connecting the Unifying Strand:

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## The Main Learning Outcomes from the Unifying Strand featured in this Unit of Learning are:

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- U.1** recall and demonstrate understanding of the fundamental concepts and procedures that underpin each strand
- U.2** apply the procedures associated with each strand accurately, effectively, and appropriately
- U.4** represent a mathematical situation in a variety of different ways, including: numerically, algebraically, graphically, physically, in words; and to interpret, analyse, and compare such representations
- U.5** make connections within and between strands
- U.6** make connections between mathematics and the real world
- U.7** make sense of a given problem, and if necessary mathematise a situation
- U.8** apply their knowledge and skills to solve a problem, including decomposing it into manageable parts and/or simplifying it using appropriate assumptions
- U.9** interpret their solution to a problem in terms of the original question
- U.13** communicate mathematics effectively: justify their reasoning, interpret their results, explain their conclusions, and use the language and notation of mathematics to express mathematical ideas precisely

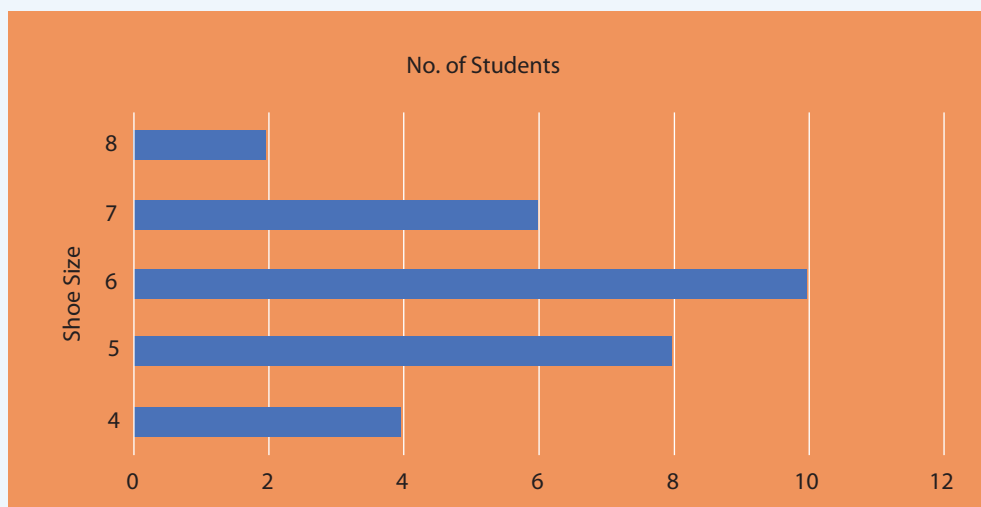
## Section C Investigations:

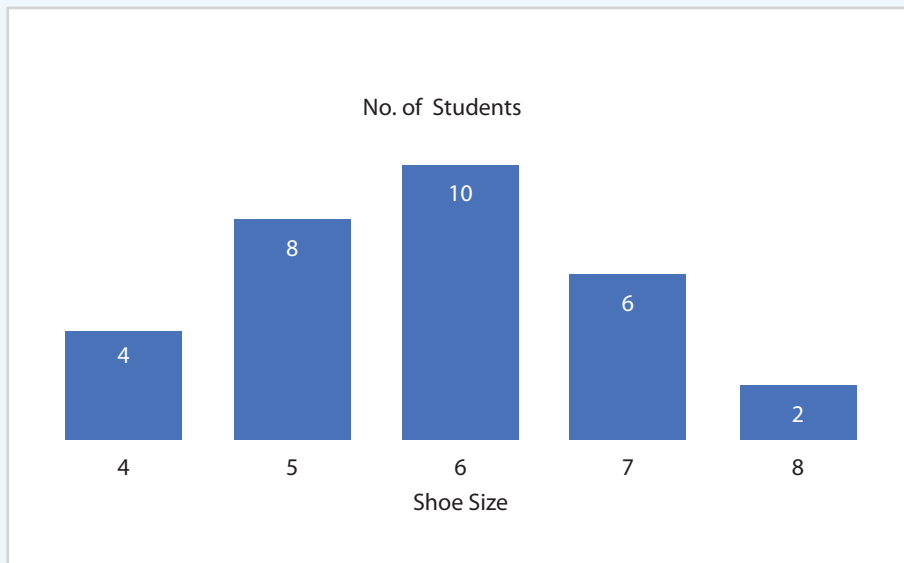
### *Investigation 1:*

Examples based on Exercise 16.2 question 6:

| Size of Shoe    | 4 | 5 | 6  | 7 | 8 |
|-----------------|---|---|----|---|---|
| No. of Students | 4 | 8 | 10 | 6 | 2 |

#### Example 1



**Example 2****Section D Learning Experiences:***Assignment 1:*

- Own student data.

See next assignment for advantages/disadvantages.

*Assignment 2:*

Some examples of advantages/disadvantages or each type:

|   | Line Plots                                                    | Bar Charts                                         | Stem and Leaf Plots                      |
|---|---------------------------------------------------------------|----------------------------------------------------|------------------------------------------|
| 1 | Best for small data sets as each piece of data requires a dot | Visually appealing                                 | Concise representation of data           |
| 2 | Only useful for small range of data                           | Can be used to compare multiple sets of data       | Shows range/max/min/clusters etc         |
| 3 | Can quickly compare data (find mode/outliers etc)             | Can be used with numerical or categorical data     | Can be used for large or small data sets |
| 4 | Exact values are retained                                     | Easy to create                                     | Not as visually appealing                |
| 5 | Easy to create                                                | Can be misleading if not started at zero frequency | Time consuming                           |

*Problem 1:*

What did the computer do on Owenahincha Beach?

1. R
2. E
3. D
4. S
5. F     IT SURFED THE NET
6. I
7. H
8. T
9. N
12. U

**Section E Chapter review:** \_\_\_\_\_

1. A line plot represents each element of the **data** by means of **dots** or other suitable symbol.
2. The bars in a bar chart can be drawn **vertically** or **horizontally**.
3. The height (or length) of each bar is a measure of the **frequency**.
4. A **key** is essential to understand the connection between the stem and the leaf in a stem and leaf plot.
5. A stem and leaf plot is used to represent **numerical** data.
6. A **line plot** can be used to represent categorical data.



# Unit of Learning:

## 17. Transformation Geometry

### Section A Unpacking the Learning Outcomes:

**GT.6:** Students should be able to investigate transformations of simple objects so that they can:

- a) recognise and draw the image of points and objects under translation, central symmetry, axial symmetry and rotation

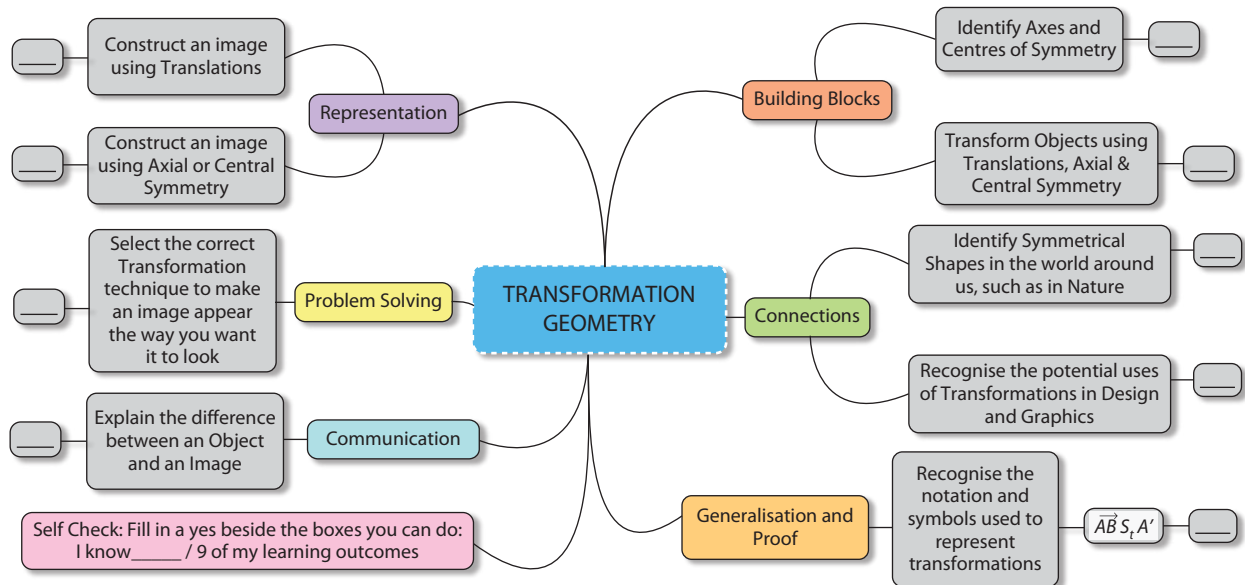
|                      |                                                                                                                                                                                                                                                                                                                               |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should:<br><ul style="list-style-type: none"> <li>• be able to identify the specified transformation</li> <li>• know the properties of the specified transformation</li> <li>• be able to interpret notations for transformations</li> </ul>                                                                         |
| <b>Skills</b>        | Students should be able:<br><ul style="list-style-type: none"> <li>• to use mathematical language and notation of transformations appropriately and accurately</li> <li>• to visualise the effect of various transformations</li> <li>• to construct points, lines and objects using the specified transformations</li> </ul> |
| <b>Understanding</b> | Students should understand:<br><ul style="list-style-type: none"> <li>• that a transformed object retains its properties following a transformation, but that the orientation may have changed</li> <li>• that objects that span an axis can still be transformed</li> </ul>                                                  |
| <b>Value</b>         | Students should:<br><ul style="list-style-type: none"> <li>• value the beauty of symmetry in patterns, both constructed and those occurring in nature</li> <li>• recognise and value the importance of correct measurement and constructions</li> </ul>                                                                       |

- b) draw the axes of symmetry in shapes

|                      |                                                                                                                                                                                                                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge</b>     | Students should know:<br><ul style="list-style-type: none"> <li>• the properties of the specified 2D shape</li> <li>• that not all shapes have an axis or axes of symmetry</li> </ul>                                                                                                                       |
| <b>Skills</b>        | Students should be able:<br><ul style="list-style-type: none"> <li>• to identify axes of symmetry in shapes</li> <li>• to draw an axis or axes of symmetry in a shape</li> </ul>                                                                                                                            |
| <b>Understanding</b> | Students should understand that:<br><ul style="list-style-type: none"> <li>• an axis of symmetry divides a shape into two congruent and symmetrical shapes</li> <li>• an axis of symmetry halves a shape, but halving the area does not necessarily mean that you have drawn an axis of symmetry</li> </ul> |
| <b>Value</b>         | Students:<br><ul style="list-style-type: none"> <li>• should value and appreciate the significance of 2D shapes in the context of the real world</li> <li>• may start to understand the connection between axes of symmetry and other real life properties such as centre of gravity</li> </ul>             |

## Section B Connecting the Unifying Strand:

All of these Learning Outcomes are summarised through the Lens of the Unifying Strand in the Mind Map, to help guide the Learning Experiences of the students:



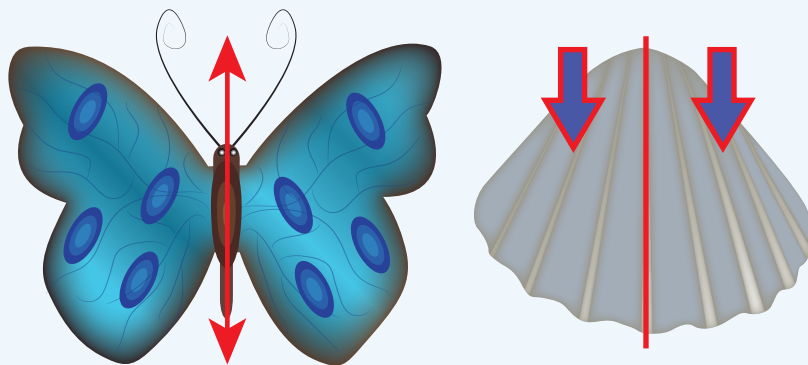
## The Main Learning Outcomes from the Unifying Strand featured in this Unit of Learning are:

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- U.9** interpret their solution to a problem in terms of the original question
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## Section C Investigations:

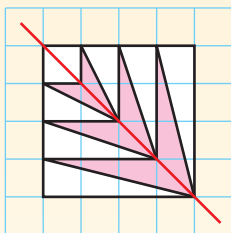
### Investigation 1: Examples:



## Section D Learning Experiences:

### Assignment 1:

(Each square represents  $5\text{ cm} \times 5\text{ cm}$ )



- i) Line of symmetry
- ii) Area of white:
 
$$5 \times 5 = 25$$

$$\frac{1}{2} \text{ of } 5 \times 10 = 25 \text{ (doubled} = 50)$$

$$\frac{1}{2} \text{ of } 5 \times 15 = 37.5 \text{ (doubled} = 75)$$

$$\frac{1}{2} \text{ of } 5 \times 20 = 50 \text{ (doubled} = 100)$$

Total(White) = 250 square cm

iii)  $\frac{250}{400} = \frac{5}{8}$

iv)  $\frac{3}{8} = 37.5\%$

v) White:Pink = 5:3

### Assignment 2:

- Triangle CED is an image of Triangle CBA by central symmetry in point C
- Triangle DEF is an image of Triangle DEC by axial symmetry in the line segment [ED]
- Triangle HID is an image of Triangle ACB by translation  $\vec{AH}$

Coordinates:

D (12, 4)

E (12, 11)

F (15, 9)

G (15, 7)

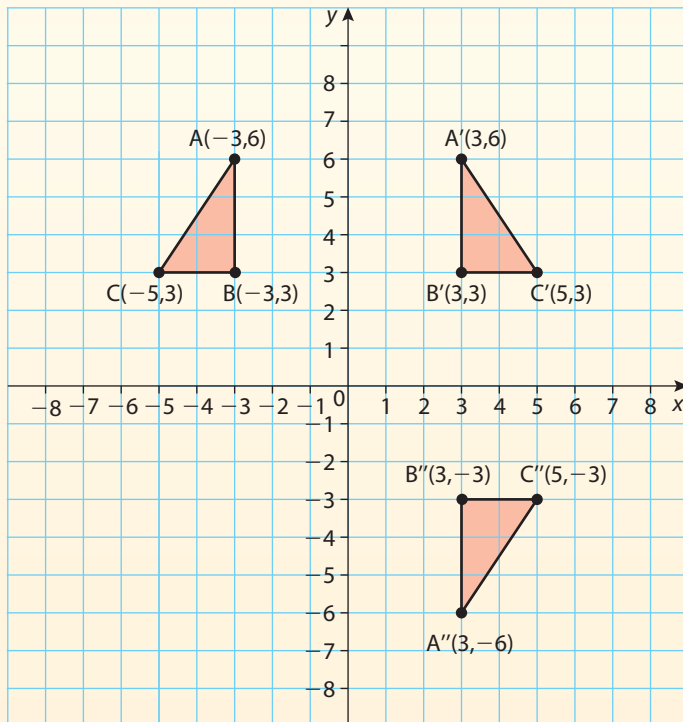
H (15, 14)

I (18, 9)

### Problem 1:

- Total area of the square is  $6 \times 6 = 36$
- If you take a  $2 \times 2$  square at the corner, the area would be 4
- By symmetry half of this has been cut away so 2 square cm have been cut from each corner
- $4 \times 2 = 8$
- $36 - 8 = 28$  square cm

### Problem 2:



The final image could have been achieved by using central symmetry through the origin.

**Section E Chapter review:**

1. In transformations **images** of objects are formed.
2. A **translation** creates an image by moving the object a fixed distance in a particular direction.
3. Axial symmetry **reflects** an object in a **line**.
4. A **central** symmetry reflects an object in a **point**.
5. To construct an image by translation you need a **set square** and a straight edge.
6. To construct an image by central symmetry you need a **straight edge**.
7. If one half of a figure folds exactly over the other half, the figure is said to be **symmetrical**.
8. The folding line in a figure is called an **axis** of symmetry.
9. Reflection through a point is called **central** symmetry.
10. When an object is reflected in a point the image is **upside down** and **back-to-front**.
11. No **triangle** has a centre of symmetry.

# Action Verbs

| Action Verb | Students should be able to:                                                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calculate   | work out a numerical answer                                                                                                                                                              |
| Construct   | use properties of shapes and geometric results to draw accurately, using only the prescribed geometrical tools                                                                           |
| Convert     | change from one form to another                                                                                                                                                          |
| Round       | give the number in the required form (for example a multiple of 100, or a number with three significant figures) that is closest in absolute terms to a particular number                |
| Interpret   | use knowledge and understanding to explain the meaning of something in context                                                                                                           |
| Understand  | have detailed knowledge of, be able to use appropriately, and see the connections between parts                                                                                          |
| Verify      | demonstrate that a statement is true                                                                                                                                                     |
| Prove       | give a deductive argument to demonstrate that a particular statement is true, including reasons for each step in the argument                                                            |
| Discuss     | offer a considered, balanced review that includes a range of arguments, factors or hypotheses; opinions or conclusions should be presented clearly and supported by appropriate evidence |
| Compare     | give an account of the similarities and (or) differences between two (or more) items or situations, referring to both (all) of them throughout                                           |
| Mathematise | generate a mathematical representation (e.g. graph, equation, geometric figure) to describe a particular aspect of a phenomenon                                                          |
| Explain     | give a reasoned account, showing how causes lead to outcomes                                                                                                                             |
| State       | provide a concise statement with little or no supporting material                                                                                                                        |
| Estimate    | state or calculate a rough value for a particular quantity                                                                                                                               |
| Define      | [a set]: give a rule that identifies the elements of a set                                                                                                                               |
| Solve       | work out an answer or solution to                                                                                                                                                        |
| Apply       | select and use knowledge and/or skills to solve a problem in a new situation                                                                                                             |
| Justify     | give valid reasons or evidence to support an answer or conclusion                                                                                                                        |

|             |                                                                                                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evaluate    | judge the relative quality or validity of something, which may include analysing, comparing and contrasting, criticising, defending or judging                                         |
| Investigate | observe, study or make a detailed and systematic examination to establish facts and reach new conclusions                                                                              |
| Sketch      | draw a rough diagram or graph without using geometrical tools                                                                                                                          |
| Analyse     | study or examine something in detail, break down to bring out the essential elements or structure; identify parts and relationships, and to interpret information to reach conclusions |
| Classify    | group things based on common characteristics                                                                                                                                           |
| Generate    | produce or create                                                                                                                                                                      |
| Use         | apply knowledge or rules to put theory into practice                                                                                                                                   |
| Generalise  | generate a general statement based on specific instances                                                                                                                               |

# Axioms and Theorems

## Axioms:

- Axiom 1:** There is exactly one line through any given two points.
- Axiom 2:** (Ruler Axiom) The properties of the distance between points.
- Axiom 3:** (Protractor Axiom) The properties of the degree measure of an angle.
- Axiom 4:** Congruent triangles (SAS, ASA and SSS).
- Axiom 5:** (Axiom of Parallels) Given any line  $l$  and a point  $P$ , there is exactly one line through  $P$  that is parallel to  $l$ .

## Theorems:

- Theorem 1:** Vertically opposite angles are equal in measure.
- Theorem 2:** In an isosceles triangle, the angles opposite the equal sides are equal. Conversely, if two angles are equal, then the triangle is isosceles.
- Theorem 3:** If a transversal makes equal alternate angles on two lines, then the lines are parallel (and converse).
- Theorem 4:** The angles in any triangle add to  $180^\circ$ .
- Theorem 5:** Two lines are parallel if and only if, for any transversal, the corresponding angles are equal.
- Theorem 6:** Each exterior angle of a triangle is equal to the sum of the interior opposite angles.
- Theorem 7:** In triangle exterior angle equals sum of two interior opposite angles.
- Theorem 8:** Two sides of a triangle are together greater than the third.
- Theorem 9:** In a parallelogram, opposite sides are equal and opposite angles are equal (and converses).
- Theorem 10:** The diagonals of a parallelogram bisect each other.
- Theorem 11:** If three parallel lines cut off equal segments on some transversal line, then they will cut off equal segments on any other transversal.
- Theorem 12:** Let  $ABC$  be a triangle. If a line  $l$  is parallel to  $BC$  and cuts  $[AB]$  in the ratio  $s:t$ , then it also cuts  $[AC]$  in the same ratio (and converse).



**Theorem 13:** If two triangles are similar, then their sides are proportional, in order (and converse) [statements only at OL].

**Theorem 14:** [Theorem of Pythagoras] In a right-angled triangle, the square of the hypotenuse is the sum of the squares of the other two sides.

**Theorem 15:** If the square of one side of a triangle is the sum of the squares of the other two sides, then the angle opposite the first side is a right angle.



Junior Cycle Maths

# Text & Tests

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